Constructed Response (Descriptive) Questions

Very Short Answer Type Questions

10 (i) Which acid is found in lemon and tomato?

(ii) The colour of pH paper changed to light green when a drop of compound is put on it. What will be the colour of pH paper when a pinch of common salt dissolved in a compound?

11 (i) What do all acids and bases in common?

(ii) What is observed when carbon dioxide gas is passed through lime water?

12 (i) Why sulphuric acid is a strong acid and acetic acid is a weak acid?

(ii) Write the name of the product formed when zinc react with sodium hydroxide?

13 Write two point of differences between washing soda and baking soda.

14 Name the acid present in ant sting and give its chemical formula. Also, give the common method to get relief from the discomfort caused by the ant sting.

Competency Based Que.

Short Answer Type Questions

15 What happens when an acid or a base is added to the water? Why does the beaker appear warm? Why should we always add acid or base to the water and not water to the acid or base?

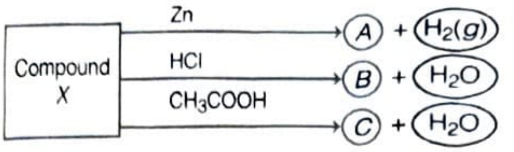
16 Give reasons:

(i) Use of a mild base like baking soda provides relief on the area stung by honeybee.

(ii) Baking powder is added to make the cakes spongy and soft.

(iii) The colour of blue copper sulphate crystals changes to white on heating. **Competency Based Que.**

17 Identify the compound X on the basis of the reactions given below. Also, write the name and chemical formula of A, B and C.



18 Answer the following: **Competency Based Que.**

(i) What happens when a concentrated solution of sodium chloride is electrolysed? Write the equation of the reaction involved.

(ii) Why is the electrolysis of a concentrated solution of sodium chloride known as chlor-alkali process?

Long Answer Type Questions

19 (i) What are neutralisation reaction?

(ii) What do you mean by strength of an acid or a base?

(iii) How pH change leads to tooth decay?

20 State in brief the method of preparation of bleaching powder. Write a balanced chemical equation for the reaction involved and state the uses of bleaching powder.

21 (i) Name the acid present in orange, tamarind and curd.

(ii) How red cabbage juice change colour in acidic and basic medium?

(iii) What do you mean by olfactory indicators? Give two examples. **Competency Based Que**

CHAPTER 03

Metals and Non-Metals

All elements and their large number of compounds are very important in our everyday life. At present, about 118 elements are known. On the basis of their properties, all of them can be divided into two main groups, i.e. **metals and non-metals**. Apart from these, some elements show properties of both metals and non-metals. These are called **metalloids.**

3.1 Metals

Those elements which form positive ions by losing electrons are called metals. e.g. Copper, iron, aluminium, sodium etc.

Physical Properties of Metals

The various physical properties of metals are as follows

(i) **Metallic lustre** In pure state, metals have a shining surface. This property is called metallic lustre. e.g. Gold, silver and platinum.

(ii) **Physical state** Generally metals exist as solids at room temperature.

Exception Mercury is the only metal which is liquid at room temperature.

(iii) **Hardness** Most of the metals are hard. e.g. Iron, copper and tin etc.

Exception Some metals like lithium, sodium and potassium are so soft that they can be easily cut with a knife. They have low densities and low melting point.

(iv) **Ductility** It is the property due to which a metal can be drawn into thin wires. Metals are generally ductile. Gold is the most ductile metal.

(v) **Malleability** It is the property of metal due to which it can be beaten into thin sheets. Most of the metals are malleable. Gold and silver are the most malleable metals.

Note It is because of their malleability and ductility that metals can be given different shapes according to our needs.

(vi) **Electrical conductivity** It is the measure of the ability of a substance to allow the flow of electric current.

Most of the metals are good conductors of electricity in solid state. However, conductivity may vary from one metal to another.

Silver is the best conductor of electricity followed by copper, gold, aluminium, tungsten, etc.

The conduction of electricity or flow of electric current occurs due to the flow of free electrons present in the metal.

Exception Mercury is a bad conductor of electricity while lead is almost non-conducting.

(vii) **Thermal conductivity** Generally, metals are good conductors of heat. e.g. Copper and silver.

Aluminium and copper are used to make cooking vessels as they are good conductors of heat.

Exception Lead and mercury are comparatively poor conductors of heat.

(viii) **Melting and boiling points** Metals generally have high melting and boiling points. Tungsten has the highest melting point among metals.

Exception Gallium and caesium have very low melting points. These two metals will melt even if we keep them on our palm.

(ix) **Sonority** The metals that produce a sound on striking a hard surface are said to be sonorous. e.g. school bells are made up of metals because of this property.

Chemical Properties of Metals

The different chemical properties of metals are as follows

1. Reaction of Metals with Oxygen

(Burning in Air or Formation of Oxides)

Almost all metals combine with oxygen (or air) to form metal oxides.

$$\text{Metal} + \text{Oxygen} \longrightarrow \text{Metal oxide}$$

e.g. $$\underset{\text{Aluminium}}{4Al(s)} + \underset{\text{Oxygen}}{3O_2(g)} \xrightarrow{\Delta} \underset{\text{Aluminium oxide}}{2Al_2O_3(s)}$$

$$2Cu(s) + O_2(g) \xrightarrow{\Delta} 2CuO(s)$$
Copper Oxygen Copper (II) oxide (Black)

Generally, metal oxides are basic in nature.

Exception Some metal oxides such as aluminium oxide, zinc oxide show both acidic and basic behaviour. Such metal oxides which react with both acids as well as bases to produce salt and water are called **amphoteric oxides.**

e.g. Aluminium oxide reacts with acids and bases in the following manner.

$$\underset{\text{(Act as basic oxide)}}{Al_2O_3(s)} + \underset{\text{Hydrochloric acid}}{6HCl(aq)} \longrightarrow \underset{\text{Aluminium chloride}}{2AlCl_3(aq)} + \underset{\text{Water}}{3H_2O(l)}$$

$$\underset{\text{(Act as acidic oxide)}}{Al_2O_3(s)} + \underset{\text{Sodium hydroxide}}{2NaOH(aq)} \longrightarrow \underset{\text{Sodium aluminate}}{2NaAlO_2(aq)} + \underset{\text{Water}}{H_2O(l)}$$

Metallic oxides are insoluble in water but some of these dissolve in water to form hydroxides known as **alkalis.**

e.g.
$$\underset{\text{Sodium oxide}}{Na_2O(s)} + \underset{\text{Water}}{H_2O(l)} \longrightarrow \underset{\text{Sodium hydroxide (Alkali)}}{2NaOH(aq)}$$

$$\underset{\text{Potassium oxide}}{K_2O(s)} + \underset{\text{Water}}{H_2O(l)} \longrightarrow \underset{\text{Potassium hydroxide (Alkali)}}{2KOH(aq)}$$

Note Alkalis are the bases that dissolve in water. The aqueous solution of a metallic (basic) oxide turns red litmus solution blue.

Order of Reactivity of Metals with Oxygen

All metals react with oxygen at different rate e.g.

- Sodium (Na) and potassium (K) react so vigorously with oxygen that they catch fire if left in the open. These are the most reactive metals. Therefore, to prevent accidental fires, these metals are kept immersed in kerosene oil.
- At room temperature, the surfaces of metals such as magnesium, aluminium, zinc, lead, etc., are covered with a thin layer of oxide which prevents the metal from further oxidation.

  Magnesium (Mg) and aluminium burns in air only by heating.
- Zinc (Zn) burns only on strong heating.
- Iron (Fe) does not burn in the form of rod or block but iron fillings burn vigorously when sprinkled in the flame of burner.
- Copper (Cu) does not burn on heating but, hot metal is coated with a black coloured layer of copper (II) oxide.
- Silver and gold do not react with oxygen even at high temperatures.

  Hence, the order of reactivity of these metals with oxygen is

  $$Na > Mg > Zn > Fe > Cu > Ag$$

Anodising

It is a process of forming a thick oxide layer of aluminium. Aluminium develops a thin layer of oxide, when left in air. This oxide layer is protective and prevents the metal from further oxidation. This layer can be made more thick by anodising. In this process, clean aluminium article is taken as anode and dil. H_2SO_4 as an electrolyte.

When electric current is passed, O_2 gas get liberated at anode, which reacts with metal aluminium to form a thicker layer of oxide on its surface. This oxide layer can be dyed easily to give aluminium articles an attractive finish.

2. Reaction of Metals with Water

Metals react with water to produce a metal oxide and hydrogen gas. Metal oxides that are soluble in water dissolve in it to further form metal hydroxide.

$$\text{Metal} + \text{Water} \longrightarrow \text{Metal oxide} + \text{Hydrogen gas}$$
$$\text{Metal oxide} + \text{Water} \longrightarrow \text{Metal hydroxide}$$

- Metals like potassium and sodium react violently with cold water. In case of sodium and potassium, the reaction is very violent and exothermic.

  $$\underset{\text{Potassium}}{2K(s)} + \underset{\text{(Cold)}}{2H_2O(l)} \longrightarrow \underset{\text{Potassium hydroxide}}{2KOH(aq)} + \underset{\text{Hydrogen}}{H_2(g)} + \text{Heat}$$

  $$\underset{\text{Sodium}}{2Na(s)} + \underset{\text{(Cold)}}{2H_2O(l)} \longrightarrow \underset{\text{Sodium hydroxide}}{2NaOH(aq)} + \underset{\text{Hydrogen}}{H_2(g)} + \text{Heat}$$

  The heat evolved is sufficient for hydrogen to catch fire. That's why, Na and K catch fire when kept in water. Therefore, both these metals are kept in 'kerosene' in order to avoid contact with both air and water.
- The reaction of calcium with water is less violent. The heat evolved is not sufficient for the hydrogen to catch fire.

  $$\underset{\text{Calcium}}{Ca(s)} + \underset{\text{Water (Room temperature)}}{2H_2O(l)} \longrightarrow \underset{\text{Calcium hydroxide}}{Ca(OH)_2(aq)} + \underset{\text{Hydrogen}}{H_2(g)}$$

  Calcium (Ca) floats over water because the bubbles of hydrogen gas formed, stick to the surface of the metal.
- Magnesium does not react with cold water. Instead, it reacts with hot water to form magnesium hydroxide and hydrogen gas. Magnesium also floats on the surface of the water as hydrogen gas bubbles produced stick to the surface of the metal.

  $$Mg(s) + 2H_2O(l) \longrightarrow Mg(OH)_2(aq) + H_2(g)$$

  Magnesium also reacts with steam to give magnesium oxide and hydrogen gas.

  $$Mg(s) + H_2O(g) \longrightarrow \underset{\text{Magnesium oxide}}{MgO(s)} + H_2(g)$$

- Metals like aluminium, iron and zinc do not react either with cold or hot water. They react with steam and form the metal oxide and hydrogen.

$$\underset{\text{Aluminium}}{2Al(s)} + \underset{\text{(Steam)}}{3H_2O(g)} \longrightarrow \underset{\text{Aluminium oxide}}{Al_2O_3(s)} + \underset{\text{Hydrogen}}{3H_2(g)}$$

$$\underset{\text{Iron}}{3Fe(s)} + \underset{\text{(Steam)}}{4H_2O(g)} \longrightarrow \underset{\text{Ferrous oxide}}{Fe_3O_4(s)} + \underset{\text{Hydrogen}}{4H_2(g)}$$

- Lead, copper, silver and gold do not react with water at all. Thus, the reactivity order of metals towards water is
  K > Na > Ca > Mg > Al > Fe > Pb > Cu > Ag > Au

3. Reaction of Metals with Acids

(i) **Reaction of metals with dil. HCl** Metals react with dilute hydrochloric acid to produce salt and hydrogen gas.

Metal + Dilute acid $\longrightarrow$ Salt + Hydrogen

e.g. $$\underset{\text{Zinc}}{Zn(s)} + \underset{\text{(Dil.)}}{2HCl(aq)} \longrightarrow \underset{\text{Zinc chloride}}{ZnCl_2(aq)} + \underset{\text{Hydrogen gas}}{H_2(g)}$$

$$\underset{\text{Iron}}{Fe(s)} + \underset{\text{(Dil.)}}{2HCl(aq)} \longrightarrow \underset{\text{Iron (II) chloride}}{FeCl_2(aq)} + \underset{\text{Hydrogen}}{H_2(g)}$$

The rate of formation of hydrogen gas bubbles decreases in the order Mg > Al > Zn > Fe. This shows the decreasing chemical reactivity of the given metals with dilute hydrochloric acid. However, copper does not reacts with dilute HCl.

(ii) **Reaction of metals with dil.** HNO_3 Hydrogen gas is not evolved when a metal reacts with nitric acid. This is due to strong oxidising nature of nitric acid. It oxidises the H_2 produced to water and itself get reduced to any of the nitrogen oxides (N_2O, NO, NO_2).

$$\text{Metal} + \underset{\text{(Dil.)}}{HNO_3} \longrightarrow \text{Metal salt} + NO/NO_2/N_2O\uparrow + H_2O$$

Exceptional case Magnesium (Mg) and manganese (Mn) react with very dil. HNO_3 to evolve H_2 gas.

$$\text{Metal (Mn/Mg)} + \underset{\text{(Dil.)}}{HNO_3} \longrightarrow \text{Salt} + H_2\uparrow$$

> ***Aqua-regia*** **(Latin for 'Royal Water')**
>
> It is a freshly prepared mixture of concentrated hydrochloric acid and concentrated nitric acid in the ratio of 3 : 1. It can dissolve gold, even though neither of these acids can do so alone. *Aqua-regia* is a highly corrosive, fuming liquid. It is one of the few reagents that is able to dissolve gold and platinum.

4. Reaction of Metals with Solutions of other Metal Salts

Reactive metals can displace less reactive metals from their compounds in solution or molten form.

If metal *A* is more reactive than metal *B* then it displaces metal *B* from its solution as follows :

Metal *A* + Salt solution of *B* $\longrightarrow$ Salt solution of *A* + Metal *B*

e.g. $$\underset{\text{Copper}}{Cu(s)} + \underset{\text{Silver nitrate}}{2AgNO_3(aq)} \longrightarrow \underset{\text{Copper nitrate}}{Cu(NO_3)_2(aq)} + \underset{\text{Silver}}{2Ag(s)}$$

This type of reaction is called **displacement reaction.**

3.2 The Reactivity Series of Metals

In general, chemical reactivity of metals is characterised by their tendency to lose electron and formation of cation. The more reactive metal will have greater tendency to lose electron. Hence, the reactivity series or activity series is a list of metals arranged in order of their decreasing activities.

The metals that are placed above hydrogen are called **most reactive metals** (e.g. K, Na, Ca etc.) and the metals that are placed below hydrogen are called **least reactive metals** (e.g. noble metals, i.e. gold and platinum).

Activity Series : Relative Reactivities of Metals

These metals are more reactive than hydrogen	K	Potassium	(Most reactive metal)
	Na	Sodium	
	Ca	Calcium	
	Mg	Magnesium	
	Al	Aluminium	
	Zn	Zinc	Reactivity decreases
	Fe	Iron	
	Pb	Lead	
	H	Hydrogen	
These metals are less reactive than hydrogen	Cu	Copper	
	Hg	Mercury	
	Ag	Silver	
	Au	Gold	(Least reactive metal)

Note Hydrogen also have non-metallic properties, but due to its electropositive nature, it has been placed in the reactivity series.

Try These 3.1

1. Some metals melt on keeping them on palm. Why? Also give an example.
2. An element *X* is soft and can be cut with a knife easily. This is very reactive with air and cannot be kept open with air. It reacts vigorously with water. Name the element *X*.
3. Why does formation of hydrogen gas faster in Al as compared to Fe when react with HCl?
4. Write a chemical reaction of metals with dil. HCl.
5. Name the reagent which is able to dissolve gold and platinum.
6. Does copper displaces silver from silver nitrate?

3.3 Non-Metals

Those elements which form negative ions by gaining electrons are called non-metals. e.g. Iodine, sulphur, oxygen, hydrogen etc.

Physical Properties of Non-Metals

The various physical properties of non-metals are as follows

(i) **Lustre** Non-metals do not have lustre.

Exception Diamond, graphite (the allotropic forms of carbon) and iodine have lustre, even though they are non-metals.

(ii) **Physical state** Non-metals are either solids or gases.

Exception Bromine is liquid at room temperature.

(iii) **Softness** Most of the non-metals are soft (if solid).

Exception Only diamond, an allotropic form of carbon is the hardest known substance.

(iv) **Malleability and ductility** Non-metals are neither malleable nor ductile.

(v) **Brittleness** Non-metals are brittle in nature. For instance, sulphur is a brittle solid. If it is hammered, it breaks into pieces.

(vi) **Electrical and thermal conductivity** Non-metals are generally poor conductors of heat and electricity.

Exception Graphite, an allotrope of carbon, is a good conductor of electricity.

(vii) **Melting and boiling points** Generally, non-metals have low melting and boiling points. But non-metals that are solids have comparatively higher boiling points, e.g. B, Si, C etc.

Note The gases like nitrogen, oxygen, carbon dioxide etc., which are constitute of air are all poor conductors of electricity.

Chemical Properties of Non-Metals

Non-metals do not react with water, steam or dilute acids to evolve hydrogen gas.

On heating, they readily form oxides or salts with conc. acids.

$$S(s) + 2H_2SO_4\text{(conc.)} \xrightarrow{\text{Heat}} 3SO_2(g) + 2H_2O(l)$$

$$S(s) + 6HNO_3\text{(conc.)} \xrightarrow{\text{Heat}} H_2SO_4(aq) + \underset{\text{(Reddish brown vapour)}}{6NO_2(g)} + 2H_2O(aq)$$

Non-metals also show displacement reaction like metals.

e.g. $\underset{\text{Chlorine}}{Cl_2(g)} + \underset{\text{Sodium bromide}}{2NaBr(l)} \longrightarrow \underset{\text{Sodium chloride}}{2NaCl(l)} + \underset{\text{Bromine}}{Br_2(g)}$

Note Most of the non-metal produces acidic oxides when dissolved in water.

Try These 3.2

1. **Do non-metals possess lustre? Give two exceptions of this property of non-metal.**
2. **Which non-metal is liquid at room temperature?**
3. **Give an example of a non-metal which is**
   (i) hardest known substance.
   (ii) a good conductor of heat and electricity.
4. **Why non-metals are brittle in nature?**
5. **Non-metals do not evolve hydrogen gas when react with water, steam or dilute acids. Why?**
6. **Write the product by giving balanced chemical equation when sulphur reacts with conc. nitric acid.**

3.4 Reaction between Metals and Non-Metals

Ionic Bond Formation

Reactivity of elements is a tendency to attain a completely filled valence shell. Thus, each element wants to have a completely filled valence shell, i.e. either 2 or 8 electrons in their outermost shell.

Metals have a tendency to loose electrons to form cations (+ve ions) and non-metals have a tendency to gain electrons to form **anions** (– ve ions).

When metals and non-metals react with each other then both of them tries to achieve completely filled outermost shell by the transfer of electrons.

This type of chemical bond formed by the complete transfer of electrons from one atom to another is called **ionic bond** or **electrovalent bond.**

The compounds formed in this manner by the transfer of electrons from a metal to a non-metal are known as ionic compounds or electrovalent compounds.

e.g.

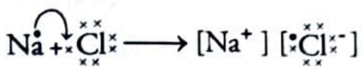
(i) **Formation of sodium chloride** Electronic configuration of **sodium** (which is a metal) is $\overset{K,L}{2,8}$... In order to complete its octet, (i.e. 8 electrons in the outermost shell), it is easier for sodium to loose one electron from the M-shell and form a cation, i.e. Na^+. The nucleus of this atom still has 11 protons but the number of electrons become 10.

$$\underset{\substack{(2,8,1)\\ \text{(Sodium atom)}}}{Na} \longrightarrow \underset{\substack{(2,8)\\ \text{(Sodium cation)}}}{Na^+} + e^-$$

Similarly, the electronic configuration of **chlorine,** (which is a non-metral) $\overset{K}{2}, \overset{L}{8}, \overset{M}{7}$. We found that it is easier for chlorine to gain one electron to complete its octet. Chlorine atom gets a unit negative charge, because its nucleus has 17 protons and 18 electrons in its K, L and M-shells.

$$\underset{\substack{(2,8,7)\\ \text{Chlorine atom}}}{Cl} + e^- \longrightarrow \underset{\substack{(2,8,8)\\ \text{(Chloride anion)}}}{Cl^-}$$

If sodium and chlorine reacts with each other then electron lost by sodium (Na^+) is gained by chlorine (Cl^-). Na^+ and Cl^- ions, being oppositely charged, attract each other and are held by strong electrostatic forces of attraction to exist as NaCl. Thus, an **ionic bond** is formed between them.



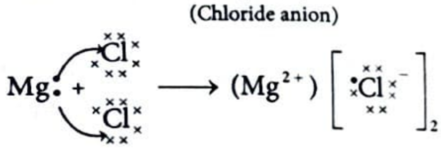
or Na^+Cl^- or NaCl

Hence, it is also clear that ionic compounds (like sodium chloride) do not exist as discrete molecules but indeed they are the aggregates of oppositely charged ions.

(ii) **Formation of magnesium chloride**

$$\underset{(2,8,2)}{Mg} \longrightarrow \underset{\substack{(2,8)\\ \text{(Magnesium cation)}}}{Mg^{2+}} + 2e^-$$

$$\underset{(2,8,7)}{Cl} + e^- \longrightarrow \underset{\substack{(2,8,8)\\ \text{(Chloride anion)}}}{Cl^-}$$



Properties of Ionic Compounds

Some important properties of ionic compounds are as follows

(i) **Physical nature** Ionic compounds are hard crystalline solids because of the strong forces of attraction between the positive and negative ions. These compounds are generally brittle and break into pieces when pressure is applied.

(ii) **Melting and boiling points** These compounds have high melting and boiling points as large amount of energy is required to break strong inter-ionic attraction.

(iii) **Solubility** Electrovalent compounds are soluble in water and insoluble in organic solvents like kerosene, benzene, ether, petrol etc.

(iv) **Conduction of electricity** Ionic or electrovalent compounds are good conductors of electricity, but they conduct electricity either in molten form or in their aqueous solution. A solution of an ionic compound in water contain ions, which move to the oppositely charged electrode when electricity is passed through it.

In molten form, the electrostatic forces of attraction between oppositely charged ions are overcome due to heat. This is the reason due to which ions move freely and conduct electricity.

They do not conduct electricity in solid form because movement of ions in the solid state is not possible due to their rigid structure.

Try These 3.3

1. **A non-metal gains electrons to form anions. What do you mean by this statement?**
2. **What type of bond is formed when electron is transfer completely from one atom to another?**
3. **Ionic compounds are crystalline solids and brittle. Why?**
4. **Why melting and boiling point of ionic compounds are high?**
5. **Water helps in separation of oppositely charged ions from their ionic compound. Give reason.**
6. **Why ions move freely and conduct electricity in the molten form?**

3.5 Occurrence of Metals

The earth's crust is the major source of metals. Sea water also contains soluble salts like sodium chloride, magnesium chloride, etc.

The elements or compounds, which occur naturally in the earth's crust are known as **minerals.** Those minerals from which metals can be extracted profitably are called **ores.**

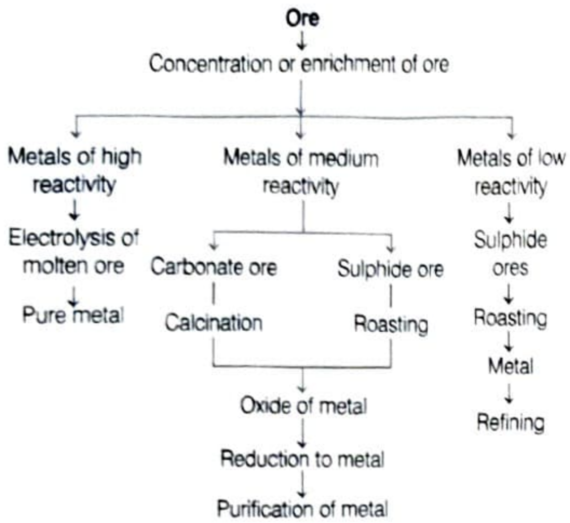
Extraction of Metals : Metallurgy

Metals have important roles in our daily life. Some metals are found in the earth's crust in the free state and some are found in the form of their compounds. The ores of many metals are their oxides. So, they have to be extracted from their ores.

The processes or various steps of obtaining pure metal from its ore is called **extraction of metals** or metallurgy.

The process used to extract metal from its ore depends on the nature of ore, nature of impurities and nature of metal to be extracted.

Several steps involved in the extraction of pure metal from their ores is given below in the form of flow chart followed by their detailed description:



However, there are some common steps which are more or less used in all metallurgical operations. These steps are follows

- Enrichment or concentration of ore.
- Enrichment of metal from concentrated ores.
- Refining of the impure metal.

Enrichment (or Concentration) of Ores

The undesirable impurities like soil, sand, etc. found in ore are called **gangue** or **matrix**.

The processes used for the removing the gangue from an ore are based on the differences between the physical and chemical properties of the gangue and the ore.

Depending upon the nature of impurities, different separation techniques for enrichment of ores are employed like gravity separation, hand picking, magnetic separation etc. Detailed study about these methods is covered in higher classes.

Extraction of Metals from Concentrated Ores

This is a process of reduction of the metal compound present in the ore.

Thus, the different techniques used for extraction of metals depend upon their position in the activity series and is divided into three categories.

- Metals of low reactivity – Metals present at the bottom of activity series.
- Metals of medium reactivity – Metals present at the middle of activity series.
- Metals of high reactivity – Metals present at the top of activity series.

Activity series and related metallurgy process are shown below

Metal	Method of extraction
K, Na, Ca, Mg, Al	Electrolysis of molten chloride or oxide
Zn, Fe, Pb, Cu	Reduction of oxide with carbon
Ag, Au, Pt	Found in native state

Belonging to the above three categories, different methods are used for extracting metals, which are explained below.

Extraction of Metals of Low Reactivity

(Metals present at the bottom of the activity series)

The metals placed at the bottom of the reactivity series are very less reactive.

Few metals are comparatively more reactive thus, they occur in their combined form and can be converted into metallic form by heating their ore in presence of oxygen. e.g.

(i) **Cinnabar** (HgS) It is an ore of mercury. When heated in air, it first changes into its oxide, i.e. HgO and then reduced to mercury metal on further heating.

e.g. $$\underset{\text{Cinnabar}}{2HgS(s)} + \underset{\text{(From air)}}{3O_2(g)} \xrightarrow{\Delta} \underset{\text{Mercuric oxide}}{2HgO(s)} + \underset{\text{Sulphur dioxide}}{2SO_2(g)}$$

$$\underset{\text{Mercury (II) oxide}}{2HgO(s)} \xrightarrow{\Delta} \underset{\text{Mercury metal}}{2Hg(l)} + \underset{\text{Oxygen}}{O_2(g)}$$

(ii) **Copper glance** (Cu_2S) It is an ore of copper, when heated in air partially, it gets oxidised and then the product reacts with the remaining copper glance to give copper metal.

e.g. $$\underset{\text{Copper glance}}{2Cu_2S(s)} + \underset{\text{(From air)}}{3O_2(g)} \xrightarrow{\Delta} \underset{\text{Copper oxide}}{2Cu_2O(s)} + \underset{\text{Sulphur dioxide}}{2SO_2(g)}$$

$$\underset{\text{Copper oxide}}{2Cu_2O(s)} + \underset{\text{Copper sulphide}}{Cu_2S(s)} \xrightarrow{\Delta} \underset{\text{Copper metal}}{6Cu(s)} + \underset{\text{Sulphur dioxide}}{SO_2(g)}$$

Extraction of Metals of Medium Reactivity

(Metals present at the middle of the activity series)

The metals in the middle of the activity series such as iron, zinc, lead, copper etc., are moderately reactive. These metals are usually present as sulphides or carbonates in nature. These sulphides or carbonates are first converted into metal oxides because it is easy to extract metals from its oxide.

Sulphides are converted into oxides by the process of roasting and carbonates are converted into oxides by the process of calcination.

The metal oxides thus obtained are then reduced to the corresponding metals by reduction using suitable reducing agents such as carbon.

Chemical reactions involved in the roasting and calcination of zinc ores followed by reduction into metals are given below.

(i) **Roasting** It is the process in which a sulphide ore is heated below its melting point in the presence of excess air to convert it into metal oxide.

e.g. $$\underset{\text{Zinc sulphide}}{2ZnS(s)} + \underset{\text{Oxygen (from air)}}{3O_2(g)} \xrightarrow{\Delta} \underset{\text{Zinc oxide}}{2ZnO(s)} + \underset{\text{Sulphur dioxide}}{2SO_2(g)}$$

(ii) **Calcination** It is a process in which a carbonate ore is heated below its melting point in the absence or limited supply of air to convert it into metal oxide.

e.g. $$\underset{\text{Zinc carbonate}}{ZnCO_3(s)} \xrightarrow{\Delta} \underset{\text{Zinc oxide}}{ZnO(s)} + \underset{\text{Carbon dioxide}}{CO_2(g)}$$

(iii) **Reduction of oxide ore** It is the process of conversion of metal oxide ore into corresponding metals. It can be done by heating the oxides with suitable reducing agents like carbon in the form of coke.

$$\underset{\text{Zinc oxide}}{ZnO(s)} + \underset{\text{Coke}}{C(s)} \longrightarrow \underset{\text{Zinc}}{Zn(s)} + \underset{\text{Carbon monoxide}}{CO(g)}$$

Sometimes displacement reactions can also be used to reduce metal oxides to metals. The highly reactive metals such as sodium, calcium, aluminium etc., are used as reducing agents because they can displace metals of lower reactivity from their compounds, e.g. Reaction of manganese dioxide with aluminium powder.

$$\underset{\text{Manganese dioxide}}{3MnO_2(s)} + \underset{\text{Aluminium}}{4Al(s)} \longrightarrow \underset{\text{Manganese}}{3Mn(l)} + \underset{\text{Aluminium oxide}}{2Al_2O_3(s)} + \text{Heat}$$

These displacement reactions are highly exothermic. The amount of heat produced is so high that the metals are produced in the molten state.

The reaction of iron (III) oxide (Fe_2O_3) with aluminium to produce iron is used to join railway tracks or cracked machine parts. This process is called **thermit welding.**

$$\underset{\text{Ferric oxide}}{Fe_2O_3(s)} + \underset{\text{Aluminium}}{2Al(s)} \longrightarrow \underset{\text{Iron}}{2Fe(l)} + \underset{\text{Aluminium oxide}}{Al_2O_3(s)} + \text{Heat}$$

This reaction is called **thermit reaction.**

Extraction of Metals of High Reactivity

(Metal present at the top of the activity series)

Metals like calcium, sodium, magnesium etc., placed at the top of reactivity series are highly reactive, so they are not found in nature as free elements.

The metallic compounds at the top of the activity series cannot be reduced by carbon or any other reducing agent due to their high affinity for oxygen than carbon. Therefore, electrolytic reduction is employed for metals like Na, Mg, Ca etc.

Electrolytic reduction Sodium, magnesium and calcium are obtained by the electrolysis of their molten chlorides. Metal is deposited at the cathode (the negative electrode) and chlorine is liberated at the anode (the positive electrode).

The reactions are as follows

At cathode, $Na^+ + e^- \longrightarrow Na$ (Reduction)

At anode, $2Cl^- \longrightarrow Cl_2 + 2e^-$ (Oxidation)

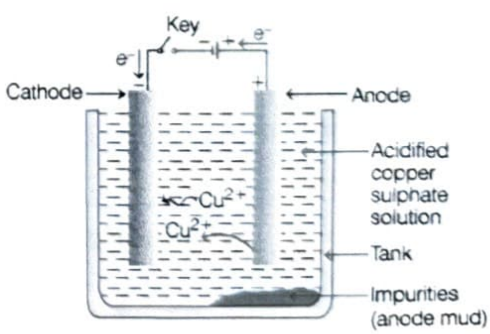
Similarly, aluminium is obtained by the electrolytic reduction of molten aluminium oxide.

Refining of Metals

It is the process of purification of the metal obtained after reduction. Various methods for refining are employed, but the most common method is the **electrolytic refining.**

Electrolytic Refining

Many metals like Cu, Zn, Ni, Ag, Au etc., are refined electrolytically. In this process, a thick block of impure metal is used as anode and a thin strip of pure metal is used as cathode. A solution of metal salt (to be refined) is used as an electrolyte. When electric current is passed, the pure metal from the anode dissolves into the electrolyte. An equivalent amount of pure metal from the electrolyte is deposited on the cathode.



Electrolytic refining of copper

This cycle is repeated until whole of the metal ion from impure block is dissolved and deposited on cathode.

The soluble impurities go into the solution whereas, the insoluble impurities settle down below anode and are known as **anode mud**.

Copper can be refined by this method. Here, anode is impure copper whereas, cathode is a strip of pure copper. The electrolyte is a solution of acidified $CuSO_4$. On passing electric current, pure copper is deposited on the cathode.

3.6 Corrosion

It is the slow process of eating away of metals by the reaction of atmospheric air and moisture, e.g. rusting of iron, tarnishing of silver, formation of green coating over copper etc.

Prevention of Corrosion

Rusting of iron is prevented by galvanising, by making alloys, painting, greasing or oiling, tin-plating and chrome plating (chromium plating).

Some of them are explained below

(i) **Galvanisation** It is the process of coating iron and steel objects with a thin layer of zinc. It is done by dipping the object in molten zinc. The galvanised article is protected against rusting even if the zinc coating is broken.

(ii) **Alloying** It is the method of improving the properties of a metal by mixing the metal with another metal or non-metal. e.g. Iron is the most widely used metal but never used in its pure state. This is because pure iron is very soft and stretches easily when hot. But, if it is mixed with a small amount of carbon (about 0.05%) then it becomes hard and strong.

Again when iron is mixed with nickel and chromium, stainless steel is obtained, which is hard and does not rust.

Alloying of gold

Pure gold is very soft. It is called **24 carat gold.** To increase strength and hardness of gold and to make it suitable for making jewellery, gold is alloyed with either silver or coppe e.g. 22 carat gold means 22 part of pure gold is alloyed with 2 parts of either Cu or Ag.

Note When the coating of metal is done with the help of electricity b making the use of other metal, it is known as **electro-plating**.

Alloy

An alloy is a homogeneous mixture of two or more metals metal and a non-metal. It is prepared by melting the prima metal and then dissolving the other elements in definite proportions. It is then cooled to room temperature.

The electrical conductivity and melting point of an alloy is le than that of pure metals, e.g. brass, (an alloy of copper and z bronze (an alloy of copper and tin) are not good conductors electricity whereas, copper is used for making electrical circ Solder (an alloy of lead and tin) has a low melting point and used for welding electrical wires together. If an alloy contai mercury as one of its components, it is called **amalgam**.

The Wonder of Ancient Indian Metallurgy

The Iron Pillar near Qutub Minar in Delhi was built more 1600 years ago by the iron workers of India. They had developed a process which prevented iron from rusting. Th due to the presence of thin layer of magnetic oxide, Fe_3O_4 surface. For its quality of rust resistance, it has been examin by scientists from all parts of the world. The Iron Pillar is 8 high and weighs 6 tonnes (6000 kg).

Try These 3.4

1. **Metal of high reactivity is extracted by which method?**
2. **How is cinnabar obtained from an ore of mercury?**
3. **What is the difference between roasting and calcination?**
4. **Which type of metals are extracted by electrolytic reductio method?**
5. **What is galvanisation? Name the element used in the pro**
6. **What is the special quality of Iron Pillar which is located n Qutub Minar?**

To Study NCERT Activities
Scan the Code

Mind Map

METALS AND NON-METALS

Non-metals
They are electronegative elements, i.e. they have a tendency to form anion by gaining electron(s) e.g. iodine (I_2), sulphur (S), hydrogen (H_2) etc.

Physical Properties
Non-metals They are non-malleable, non-ductile and bad conductor of heat and electricity. They possess low melting point and boiling point.
e.g. C, S, P, N, O, etc.

Exceptions among non-metals
(a) Non-metals are generally solids or gases except bromine which is liquid.
(b) Iodine is a lustrous non-metal.
(c) Diamond is the hardest natural known substance.

Chemical Properties
- **Reaction with oxygen**
Non-metals also form oxides but their nature is generally acidic, (e.g. P_2O_5, SO_2 and CO_2 as they produce acid with water) or neutral, (e.g. CO, H_2O, etc.)
- **Acidic oxides + Water ⟶ Acids**
$CO_2(g) + H_2O(l) \longrightarrow \underset{\text{Carbonic acid}}{H_2CO_3(aq)}$
- Non-metals do not react with water to evolve hydrogen gas.
- Non-metals do not displace H_2 from acids because non-metals are electron acceptor and they cannot supply electron to hydrogen.

Alloy
- These are homogeneous mixture of two or more metals or a metal and a non-metals.
- Alloy of metal with Hg (mercury) is called amalgam.

Metalloid
Some elements shows properties of both metals and non-metals. These are called metalloids.
eg. Si, Ge, As, Sb and Te.

Ionic or Electrovalent Bond
The bond formed by the complete transfer of an electron from a metal atom to a non-metal atom is called ionic bond.

Properties of Ionic Compounds
- They are brittle and have high melting and boiling point.
- They are soluble in water.
- They are good conductor of electricity in aqueous.
solution or in molten state because of the presence of free ions.

Physical Properties
Metals They metals are malleable, ductile and good conductor of heat and electricity. They possess high melting point and boiling point. e.g. Cu, Al, Mg, etc

Exception among metals
(a) All metals are solid but mercury is a liquid at room temperature.
(b) Metals have high melting and boiling points. But gallium & caesium have very low melting points. These two metals will melt if you keep them on your palm.

Metals
They are electropositive in nature and have a tendency to form positive ion by losing electon(s),e.g. Cu, Fe, Au, Na etc.

Reactivity series or activity series
It is a list of metals arranged in order of their decreasing activities.
The decreasing order of reactivity is
K > Na > Ca > Mg > Al > Zn > Fe > Pb > H > Cu > Hg > Ag > Au

Chemical Properties
- **Metals combine with oxygen** and form basic oxides.
$2Mg(s) + O_2(g) \longrightarrow 2MgO(s)$
Basic oxides + water ⟶ Bases
$Na_2O(s) + H_2O(l) \longrightarrow \underset{\text{Sodium hydroxide}}{2NaOH(aq)}$
In litmus test, the bases formed turn red litmus paper blue.
- **Reaction with water**
–Na and K react violently with cold water. Evolved H_2 gas catches fire.
$2K(s) + 2H_2O(l) \rightarrow 2KOH(aq) + H_2(g)$
–Magnesium reacts with hot water
$Mg(s) + \underset{\text{(Hot)}}{2H_2O(l)} \rightarrow Mg(OH)_2(aq) + 3H_2(g)$
–Al, Zn and Fe react with steam.
$2Al(s) + 3H_2O(g) \rightarrow Al_2O_3(s) + 3H_2(g)$
–Lead (Pb), Copper (Cu), Silver (Ag) and Gold (Au) do not react with water.
- **Reactions with Acids**
–Metals react with dil. HCl and H_2SO_4 to liberate H_2 gas.
Metal + dil. acid → Salt +Hydrogen.
–When metals react with dilute HNO_3, H_2 gas is not evolved but H_2O (water) is formed.
Metal + dil. nitric acid → Metal salt + Water + Oxides of nitrogen.

Metal Extraction

Metallurgy
It is the process of extraction of metals from their ores.

Minerals
The elements or compounds which occur naturally in the earth's crust are known as minerals.

Ores and Gangue
The minerals from which metal is extracted profitably are called the ores and the impurities associated with them are called gangue.

Aqua-regia
It is mixture of concentrated hydrochloric acid and concentrated nitric acid in the ratio of 3:1 Gold and platinum can dissolve in *aqua-regia*.

Galvanisation
It is the process of coating iron and steel objects with a thin layer of zinc.

Steps of Extraction

Concentration of Ore
It is the process used for the removal of impurities of sand, clay etc., from the metal.

Metals of High Reactivity → Electrolysis of molten ore → Pure metal

Metals of Medium Reactivity → Carbonate ore → Calcination → Oxide of metal; Sulphide ore → Roasting → Oxide of metal; Oxide of metal → Reduction to metal → Purification of Metal

Metals of Low Reactivity → Sulphide ores → Roasting → Metal → Refining

NCERT FOLDER

Complete Explanations of all NCERT's Questions

Intext Questions

Q.1 Give an example of a metal which

(i) is a liquid at room temperature.
(ii) can be easily cut with a knife.
(iii) is a good conductor of heat.
(iv) is a poor conductor of heat. **Page 40**

Sol. (i) Mercury (ii) Sodium (iii) Silver (iv) Lead

Q.2 Explain the meanings of malleable and ductile. **Page 40**

Sol. **Malleable** A substance or material, which can be beaten into thin sheets is called malleable, e.g. metals like Ag (silver), Au (gold) etc.

Ductile A substance which can be drawn into thin wires is called ductile, e.g. metals like Ag, Au etc.

Q.3 Why is sodium kept immersed in kerosene oil? **Page 46**

Sol. Sodium metal is highly reactive metal. It reacts so vigorously with oxygen and catches fire if kept in open air. Therefore, to protect it from accidental fires, sodium is kept immersed in kerosene oil.

Q.4 Write equations for the reactions of

(i) iron with steam.
(ii) calcium and potassium with water. **Page 46**

Sol. (i) $\underset{\text{Iron}}{3Fe(s)} + \underset{\text{Steam}}{4H_2O(g)} \longrightarrow \underset{\text{Iron oxide}}{Fe_3O_4(s)} + \underset{\text{Hydrogen}}{4H_2(g)}$

(ii) (a) $\underset{\text{Calcium}}{Ca(s)} + \underset{\text{Water}}{2H_2O(l)} \longrightarrow \underset{\text{Calcium hydroxide}}{Ca(OH)_2(aq)} + \underset{\text{Hydrogen}}{H_2(g)}$

(b) $\underset{\text{Potassium}}{2K(s)} + \underset{\text{Water}}{2H_2O(l)} \longrightarrow \underset{\text{Potassium hydroxide}}{2KOH(aq)} + \underset{\text{Hydrogen}}{H_2(g)}$

Q.5 Samples of four metals *A*, *B*, *C* and *D* were taken and added to the following solution one by one. The results obtained have been tabulated as follows

Metal	Iron (II) sulphate	Copper (II) sulphate	Zinc sulphate	Silver nitrate
A	No reaction	Displacement	–	–
B	Displacement	–	No reaction	–
C	No reaction	No reaction	No reaction	Displacement
D	No reaction	No reaction	No reaction	No reaction

Use the given table to answer the following questions about metals *A*, *B*, *C* and *D*.

(i) Which is the most reactive metal?
(ii) What would you observed if *B* is added to a solution of copper (II) sulphate?
(iii) Arrange the metals *A*, *B*, *C* and *D* in the order of decreasing reactivity. **Page 4**

Sol. (i) *B* is the most reactive metal as it displaces iron from its salt solution (iron sulphate).

(ii) *B* will displace Cu from $CuSO_4$ solution because *B* is more reactive than copper.

(iii) The order of reactivity is $B > A > C > D$.

Q.6 Which gas is produced when dilute hydrochloric acid is added to a reactive metal? Write the chemical reaction when iron reacts with dil. H **Page 4**

Sol. Hydrogen gas is produced when dilute hydrochloric acid added to a reactive metal.
The chemical reaction when iron reacts with dil. H_2SO_4 as follows

$$\underset{\text{Iron}}{Fe(s)} + \underset{\text{Dil. sulphuric acid}}{H_2SO_4(aq)} \longrightarrow \underset{\text{Ferrous sulphate}}{FeSO_4(aq)} + \underset{\text{Hydrogen}}{H_2(g)}$$

Q.7 What would you observe when zinc is added to a solution of iron (II) sulphate? Write the chemical reaction that takes place. **Page 4**

Sol. Zinc being more reactive than iron displaces iron from iron (II) sulphate solution. As the result, the green colour of the solution fades and iron metal gets deposited.
The chemical reaction when iron reacts with dil. H_2SO_4 as follows.

$$\underset{\text{Zinc}}{Zn(s)} + \underset{\text{(Green)}}{FeSO_4(aq)} \longrightarrow \underset{\text{(Colourless)}}{ZnSO_4(aq)} + \underset{\text{Iron (Deposited)}}{Fe(s)}$$

Q.8 (i) **Write the electron-dot structures for sodium, oxygen and magnesium.**
(ii) **Show the formation of Na_2O and MgO by the transfer of electrons.**
(iii) **What are the ions present in these compounds?** **Page 4**

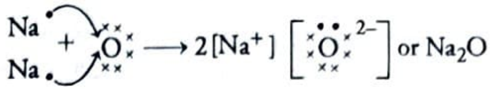
Sol. (i) The electron dot structures of sodium (Na), oxygen (O) and magnesium (Mg) are tabulated below.

Element	Atomic number	Electronic configuration	Electron dot structure
Na	11	2, 8, 1	$\dot{\text{Na}}$
O	8	2, 6	:Ö:
Mg	12	2, 8, 2	:Mg

(ii) Ionic bond is formed in Na_2O and MgO.

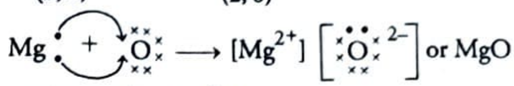
(a) Na_2O $\quad \underset{(2,8,1)}{2Na} \longrightarrow \underset{(2,8)}{2Na^+} + 2e^-$

$\underset{(2,6)}{O} + 2e^- \longrightarrow \underset{(2,8)}{O^{2-}}$



(b) MgO $\quad \underset{(2,8,2)}{Mg} \longrightarrow \underset{(2,8)}{Mg^{2+}} + 2e^-$

$\underset{(2,6)}{O} + 2e^- \longrightarrow \underset{(2,8)}{O^{2-}}$



(iii) Na_2O has Na^+ and O^{2-} ions.
MgO has Mg^{2+} and O^{2-} ions.

Q.9 Why do ionic compounds have high melting points? **Page 49**

Sol. In ionic compounds, strong electrostatic forces of attraction are present between the oppositely charged ions. When these compounds are heated, a lot of heat energy is required to break these strong electrostatic forces of attraction during melting. Therefore, ionic compounds have high melting and boiling points.

Q.10 Define the following terms.
(i) Mineral (ii) Ore (iii) Gangue **Page 53**

Sol. (i) The naturally occurring elements or compounds of metals present in the earth's crust are called **minerals.**
(ii) **Ores** are those minerals from which a particular metal can be extracted profitably by using certain process.
(iii) The undesirable impurities present in the ore are called **gangue** or **matrix.**

Q.11 Name two metals which are found in nature in the free state. **Page 53**

Sol. Gold and platinum are the two metals that are found in nature in the free state.

Q.12 What chemical process is used for obtaining a metal from its oxide? **Page 53**

Sol. Metal is obtained from its oxide by reduction. The reduction can be done either by heating with carbon (coke) or by using highly reactive metals such as sodium, calcium, aluminium etc.

e.g. $\underset{\text{Zinc oxide}}{ZnO(s)} + \underset{\text{Coke}}{C(s)} \longrightarrow \underset{\text{Zinc}}{Zn(s)} + \underset{\text{Carbon monoxide}}{CO(g)}$

This reaction is highly exothermic. The heat produced is so large that metal is produced in the molten state.

Q.13 Metallic oxides of zinc, magnesium and copper were heated with the following metals:

Metal	Zinc	Magnesium	Copper
Zinc oxide			
Magnesium oxide			
Copper oxide			

In which cases will you find displacement reactions taking place? **Page 55**

Sol. As the order of the reactivity of zinc, magnesium and copper is Mg > Zn > Cu.
Thus, zinc will react with copper oxide to displace copper. Magnesium will also displace zinc from zinc oxide and copper from copper oxide. Copper, being least reactive will not react with any of the given oxides.

Metal	Zinc	Magnesium	Copper
Zinc oxide	–	✓	–
Magnesium oxide	–	–	–
Copper oxide	✓	✓	–

Q.14 Which metals do not corrode easily? **Page 55**

Sol. Metals present at the bottom of the reactivity series do not corrode easily, e.g. gold, silver, platinum etc.

Q.15 What are alloys? **Page 55**

Sol. An alloy is a homogeneous mixture of two or more metals or a metal and a non-metal. It is prepared by mixing the metals in molten form and then cooling the mixture. The electrical conductivity and melting point of an alloy is less than that of pure metals.

e.g.

Alloy	Composition	Uses
Brass	Copper and zinc	Utensils and taps
Bronze	Copper and tin	Medals, statues and valves

Exercise

Q.1 Which of the following pairs will give displacement reactions?
(i) NaCl solution and copper metal
(ii) $MgCl_2$ solution and aluminium metal
(iii) $FeSO_4$ solution and silver metal
(iv) $AgNO_3$ solution and copper metal

Sol. *(iv)* $AgNO_3$ solution will give displacement reaction with copper (Cu) because copper is placed above silver in the activity series, i.e. copper is more reactive than silver.

$$2AgNO_3(aq) + Cu(s) \longrightarrow Cu(NO_3)_2(aq) + 2Ag(s)$$

Q.2 Which of the following methods is suitable for preventing an iron frying pan from rusting?
(i) Applying grease (ii) Applying paint
(iii) Applying a coating of zinc (iv) All of these

Sol. *(iii)* Applying a coating of zinc will prevent an iron frying pan from rusting.

Q.3 An element reacts with oxygen to give a compound with a high melting point. This compound is also soluble in water. The element is likely to be
(i) calcium (ii) carbon
(iii) silicon (iv) iron

Sol. (*i*) The compound is likely to be calcium (Ca) because it combines with oxygen to give CaO (calcium oxide) with very high melting point. Calcium oxide dissolves in water to form calcium hydroxide.

$$2Ca(s) + O_2(g) \longrightarrow 2CaO(s)$$

$$CaO(s) + H_2O(l) \longrightarrow \underset{\text{Calcium hydroxide}}{Ca(OH)_2(aq)}$$

Q.4 Food cans are coated with tin and not with zinc because

(i) zinc is costlier than tin.
(ii) zinc has a higher melting point than tin.
(iii) zinc is more reactive than tin.
(iv) zinc is less reactive than tin.

Sol. (*iii*) Food cans are not coated with zinc because being more reactive than tin, zinc can react with organic acids present in the food.

Q.5 You are given a hammer, a battery, a bulb, wires and a switch.

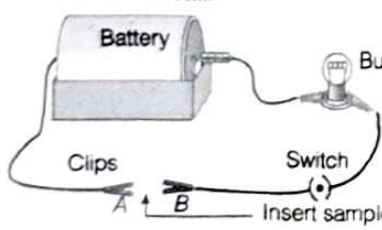
(i) How could you use them to distinguish between samples of metals and non-metals?
(ii) Assess the usefulness of these tests in distinguishing between metals and non-metals.

Sol. (i) There are some ways by which we can distinguish between metals and non-metals as follows.

(a) Take the given samples of metals and non-metals; and strike them with the hammer.
If it converts into a sheet, it is a metal and if not, it is non-metal. As metals are malleable, while non-metals are not.

(b) If it produces sound when struck with the hammer, it is a metal. But if it does not produce a sound, it is non-metal.

(c) Now arranging the given objects to form an electric circuit. Insert any one sample between clips *A* and *B*. If the bulb glows, it is a metal (good conductor of electricity). If the bulb does not glow, it is a non-metal.



Electrical circuit diagram to show conductivity of metals

(ii) From the above tests, it is clear that metals are generally malleable, sonorous and good conductors of electricity, while non-metals are generally non-malleable/brittle, non-sonorous and poor conductors of electricity.

Q.6 What are amphoteric oxides? Give two examples of amphoteric oxides.

Sol. The metallic oxides which show the properties of acids as well as bases are called amphoteric oxides. It means that they react with both bases and acids to form salt and water.
e.g. ZnO and Al_2O_3.

$$\underset{\substack{\text{Zinc oxide}\\\text{(As a base)}}}{ZnO(s)} + \underset{\substack{\text{Hydrochloric}\\\text{acid}}}{2HCl(aq)} \longrightarrow \underset{\text{Zinc chloride}}{ZnCl_2(aq)} + \underset{\text{Water}}{H_2O(l)}$$

$$\underset{\substack{\text{Zinc oxide}\\\text{(As an acid)}}}{ZnO(s)} + \underset{\text{Sodium hydroxide}}{2NaOH(aq)} \longrightarrow \underset{\text{Sodium zincate}}{Na_2ZnO_2(aq)} + \underset{\text{Water}}{H_2O(l)}$$

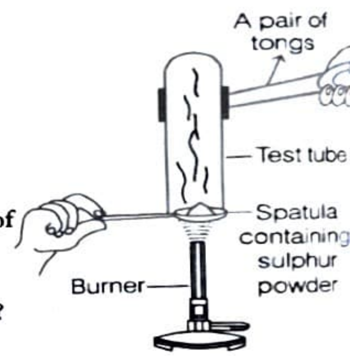
Q.7 Name two metals which will displace hydrogen from dilute acids, and two metals which will not.

Sol. Zinc and magnesium displace H_2 from dilute acids wh copper and silver do not.

Q.8 In the electrolytic refining of a metal *M*, what would y take as the anode, the cathode and the electrolyte?

Sol. **Anode** (positively charged) Block of the impure metal
Cathode (negatively charged) Strip of the pure metal
Electrolyte Aqueous solution of a salt of the metal *M*.

Q.9 Pratyush took sulphur powder on a spatula and heated it. He collected the gas evolved by inverting a test tube over it, as shown in the figure below.



Collection of gas

(i) What will be the action of the gas on ?
(a) dry blue litmus paper?
(b) moist blue litmus paper?
(ii) Write a balanced chemical equation for the reaction taking place.

Sol. (i) (a) No change in colour will take place in case of blue litmus paper.
(b) The moist blue litmus paper will change its colou red because sulphur is non-metal and so, its ox are acidic in nature.

(ii) $S(s) + O_2(g) \longrightarrow SO_2(g)$

$$SO_2(g) + H_2O(l) \longrightarrow \underset{\text{Sulphurous acid}}{H_2SO_3(aq)}$$

Q.10 State two ways to prevent the rusting of iron.

Sol. Ways to prevent rusting of iron are as follows

(i) By painting or greasing the surface of iron, it can prevented from rusting because it protects iron fr direct contact with air.
(ii) By galvanising the iron surface, a layer of zinc is co on it. Zinc is more reactive than iron, so it will be ru in preference of iron.

Q.11 What type of oxides are formed when non-metals combine with oxygen?

Sol. Non-metals form acidic oxides, i.e. their aqueous solu turns blue litmus solution red.

$$\underset{\substack{\text{Carbon}\\\text{(Non - metal)}}}{C(s)} + \underset{\text{Oxygen}}{O_2(g)} \longrightarrow \underset{\text{Carbon dioxide}}{CO_2(g)}$$

$$\underset{\substack{\text{Carbon}\\\text{dioxide}}}{CO_2(g)} + \underset{\text{Water}}{H_2O(l)} \longrightarrow \underset{\text{Carbonic acid}}{H_2CO_3(aq)} \xrightarrow[\substack{\text{litmus}\\\text{solution}}]{\text{Blue}} \text{Red}$$

Q.12 Give reasons

(i) **Platinum, gold and silver are used to make jewellery.**
(ii) **Sodium, potassium and lithium are stored under oil.**
(iii) **Aluminium is a highly reactive metal, yet it is used to make utensils for cooking.**
(iv) **Carbonate and sulphide ores are usually converted into oxides during the process of extraction.**

Sol. (i) Platinum, gold and silver are highly malleable, lustrous and least reactive, so they are not corroded by air and water easily.

(ii) Sodium, potassium and lithium are very reactive, so these metals react vigorously with atmospheric oxygen to form oxides. Storing them in oil prevents their oxidation.

(iii) A thin layer of aluminium oxide formed on the surface of aluminium prevents it from corrosion. Thus, aluminium vessels do not react with any ingredient of food and are suitable for cooking.

(iv) It is easier to obtain metal from its oxide rather than its carbonates and sulphides. Therefore, these ores are first converted to oxide and then reduced to metal.

Q.13 You must have seen tarnished copper vessels being cleaned with lemon or tamarind juice. Explain, why these sour substances are effective in cleaning the vessels?

Sol. A layer of basic copper carbonate is formed by the reaction of air on copper metal.

$$2Cu(s) + \underbrace{O_2(g) + H_2O(l) + CO_2(g)}_{\text{(From air)}} \longrightarrow \underbrace{CuCO_3 \cdot Cu(OH)_2(s)}_{\text{Basic copper carbonate}}$$

This layer being insoluble in water, cannot be cleaned with water alone. But, it is soluble in acids so lemon containing citric acid, tamarind containing tartaric acid or any other sour substance containing acid can be effective in cleaning the vessels. As these acids neutralises the basic copper carbonate and dissolves the layer. Hence, surface of copper vessels are cleaned with lemon or tamarind juice to give its characteristics lustre.

Q.14 Differentiate between metals and non-metals on the basis of their chemical properties.

Sol.

Chemical properties	Metals	Non-metals
Formation of ions	They are electropositive elements, lose electron(s) and form cations. e.g. $Na \longrightarrow Na^+ + e^-$	They are electronegative elements, gain electron(s) and form anions. $Cl + e^- \longrightarrow Cl^-$
Discharge of ions	Discharge at the cathode during electrolysis of their compound. At cathode, $\underset{\text{Cation}}{Na^+} + e^- \longrightarrow Na$	Liberated at anode during electrolysis. Except hydrogen, liberated at cathode. At cathode, $2H^+ + 2e^- \longrightarrow H_2\uparrow$
Reducing or Oxidising agent	They are reducing agents as they donate electrons during chemical reaction.	They are oxidising agents, as they accept electrons during chemical reaction.
Nature of oxides	Metallic oxides are basic. Some of them dissolves in water forming alkaline solution. e.g. Basic oxide : K_2O, Na_2O, CaO, MgO and CuO Amphoteric oxide : Al_2O_3, PbO and ZnO	Non-metallic oxides are acidic in nature. e.g. Acidic oxide : CO_2, SO_2, SO_3, NO_2 and P_2O_5 Neutral oxide : CO, NO, N_2O and H_2O
Reaction with acids	Active metals react with dil. HCl/H_2SO_4 to yield H_2 gas and salts. $M + 2HCl \longrightarrow MCl_2 + H_2\uparrow$	Non-metals do not react with dilute acids, as they cannot replace H^+ ion from an acid to form salt.

Q.15 A man went door to door posing as a goldsmith. He promised to bring back the glitter of old and dull gold ornaments. An unsuspecting lady gave a set of gold bangles to him which he dipped in a particular solution. The bangles sparkled like new but their weight was reduced drastically.

The lady was upset but after a futile argument, the man beat a hasty retreat. Can you play the detective to find out the nature of the solution he had used?

Sol. The man used *aqua-regia*, a mixture of conc. HCl and conc. HNO_3 in the ratio of 3 : 1. This is the only solution, that can dissolve gold. As the gold from the bangles was dissolved in *aqua-regia*, their weight was reduced drastically.

Q.16 Give reasons, why copper is used to make hot water tanks and not steel (an alloy of iron)?

Sol. Iron (or steel) is more reactive than copper. Iron in contact with steam form Fe_3O_4.

$$3Fe(s) + 4H_2O(g) \longrightarrow Fe_3O_4(s) + 4H_2(g)$$

So, the body of tank is made of copper but not steel as copper does not react with water.

REVIEW EXERCISE

Including Competency Based Questions

Competency Focused (Objective) Questions

Multiple Choice Questions

1 Which of the following metal has highest melting point?

(a) Copper (b) Silver (c) Sodium (d) Tungsten

Sol. (*d*) Among the given metals. Tungsten has the highest melting point

2 Which of the following is a characteristic of metals?

(a) They have one to three valence electrons.

(b) They have 4 to 8 valence electrons.

(c) They are brittle.

(d) They are capable to form anions easily.

Sol. (*a*) Metal can easily given up their electrons and form electropositive ions. They have one to three valence electrons. They are not brittle and do not form anions.

3 Which one of the following four metals would be displaced from the solution of its salts by other three metals? **NCERT Exemplar**

(a) Mg (b) Ag

(c) Zn (d) Cu

Sol. (*b*) Silver (Ag) metal would be displaced from the solution of its salts by other three metals because Ag is less reactive than Mg, Zn and Cu.

4 Which of the following reaction shows that the given oxide is amphoteric in nature?

(a) $2Zn + O_2 \xrightarrow{\Delta} 2ZnO$

(b) $ZnO + H_2SO_4 \longrightarrow ZnSO_4 + H_2O(l)$

(c) $ZnO + 2NaOH \longrightarrow Na_2ZnO_2 + H_2O(l)$

(d) (b) and (c) together **NCERT Exemplar**

Sol. (*d*) Metals when react with the oxygen, give basic-oxides/ amphoteric oxides. ZnO is an amphoteric oxide.

Option (b) and (c) indicates that, ZnO react with the acid (H_2SO_4) as well as with the base (NaOH).

Hence, (b) and (c) together shows the amphoteric nature of oxide.

5 What happens when calcium is treated with water?

(a) It does not react with water.

(b) It reacts less violently with water.

(c) Bubbles of hydrogen gas formed stick to the surface of calcium.

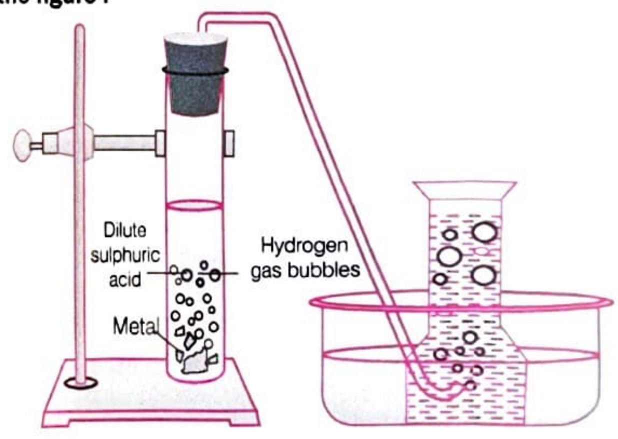
(d) Both (b) and (c)

Sol. (*d*) Calcium reacts less violently with water and the bubbles of hydrogen gas produced stick to the surface of calcium. Due to which it floats over water surface.

$$Ca(s) + 2H_2O(l) \longrightarrow \underset{\text{Calcium hydroxide}}{Ca(OH)_2(aq)} + H_2(g)\uparrow$$

Much less heat is produced in this reaction due to which hydrogen gas formed does not catch fire.

6 A reactive metal (*M*) is treated with H_2SO_4 (dil). The gas is evolved and is collected over the water as shown in the figure :



The correct conclusion drawn is **Competency Based Qu**

(a) the gas is oxygen.

(b) the gas is lighter than air.

(c) the gas is SO_2 and is lighter than air.

(d) the gas is H_2 and is heavier than air.

Sol. (*b*) When any reactive metal (*M*) reacts with the acid H_2SO_4(dil.), it evolves hydrogen gas (H_2) which is lighter than air.

$$M(s) + H_2SO_4(\text{dil.}) \longrightarrow M\text{-sulphate} + H_2(g)\uparrow$$

7 The composition of *aqua-regia* is

(a) Dil.HCl : Conc. HNO_3
3 : 1

(b) Conc. HCl : Dil. HNO_3
3 : 1

(c) Conc. HCl : Conc. HNO_3
3 : 1

(d) Dil. HCl : Dil. HNO_3
3 : 1

Sol. (*c*) Conc. HCl and conc. HNO_3 in 3 : 1 ratio form *aqua-regia*. It is a highly corrosive, fuming liquid, which can dissolve all metals even gold and platinum also.

8 Metal oxides generally react with acids, but few oxides of metal also react with bases. Among the given options, the metallic oxides which reacts with both an acid and a base is

(a) MgO (b) ZnO
(c) Al_2O_3 (d) CaO

Sol. (*c*) Among the given options, Al_2O_3 is the metallic oxide which reacts with both an acid and a base.

9 Sodium hydroxide is termed an alkali while ferric hydroxide is not because **CBSE 2023**

(a) sodium hydroxide is a strong base, while ferric hydroxide is a weak base.
(b) sodium hydroxide is a base which is soluble in water while ferric hydroxide is also a base but it is not soluble in water.
(c) sodium hydroxide is a strong base while ferric hydroxide is a strong acid.
(d) sodium hydroxide and ferric hydroxide both are strong base but the solubility of sodium hydroxide in water is comparatively higher than that of ferric hydroxide.

Sol. (*b*) Alkali are the bases that dissolve in water. Thus, sodium hydroxide is an alkali, while ferric hydroxide is not.

10 An element '*X*' reacts with O_2 to give a compound with a high melting point. This compound is also soluble in water. The element '*X*' is likely to be **CBSE 2020**

(a) iron (b) calcium
(c) carbon (d) silicon

Sol. (*b*) Calcium reacts with O_2 to give CaO, which have high melting point and it is soluble in water.

11 The arrangement for copper, tin, lead and mercury according to the reactivity series is

(a) zinc > lead > copper > mercury
(b) lead > copper > mercury > zinc
(c) copper > mercury > zinc > lead
(d) mercury > zinc > lead > copper

Sol. (*a*) According to reactivity series, the correct order is, zinc (Zn) > lead (Pb) > copper (Cu) > mercury (Hg).

12 Which of the following metal will not give $H_2(g)$ with H_2O?

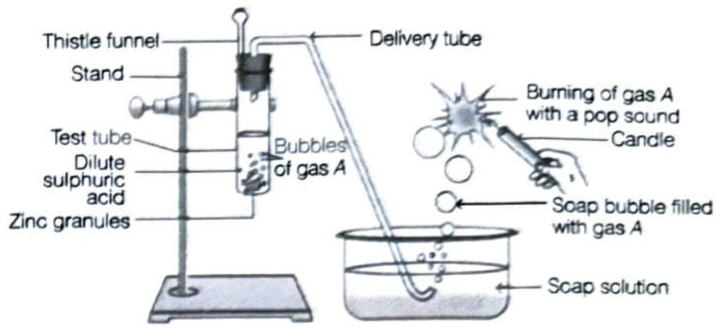
(a) $Na(s) + 2H_2O \longrightarrow$ (b) $Mg(s) + H_2O \longrightarrow$
(c) $Zn(s) + H_2O \longrightarrow$ (d) $Cu + H_2O \longrightarrow$

Sol. (*d*) Metals placed below the hydrogen in reactivity series will not give $H_2(g)$ with water (H_2O).

Decreasing order of reactivity of metals is

$$Na > Mg > Zn > Cu$$

13 Identify gas *A* in the following experiment.



Competency Based Que.

(a) Nitrogen (b) Hydrogen
(c) Oxygen (d) Carbon dioxide

Sol. (*b*) When zinc granules react with dilute sulphuric acid, then hydrogen gas is produced.

$$\underset{\text{Zinc}}{Zn} + \underset{\text{Dil. sulphuric acid}}{H_2SO_4} \longrightarrow \underset{\text{Zinc sulphate}}{ZnSO_4} + \underset{\text{Hydrogen}}{H_2 \uparrow}$$

This hydrogen makes a pop sound on burning due to the reaction between hydrogen and oxygen present in air.

14 Which of the following oxide(s) of iron would be obtained on prolonged reaction of iron with steam? **NCERT Exemplar**

(a) FeO (b) Fe_2O_3
(c) Fe_3O_4 (d) Fe_2O_3 and Fe_3O_4

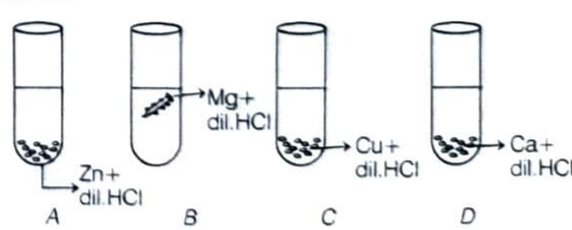
Sol. (*c*) Reactions of iron metal with water.

$$\underset{\text{Iron}}{Fe(s)} + \underset{\text{Steam}}{H_2O} \longrightarrow \underset{\text{Iron (II) oxide}}{FeO} + H_2(g)\uparrow$$

$$\underset{\text{Iron}}{2Fe(s)} + \underset{\text{Steam}}{3H_2O} \longrightarrow \underset{\text{Iron (III) oxide}}{Fe_2O_3} + 3H_2(g)\uparrow$$

$$3Fe + 4H_2O \longrightarrow \underset{\text{Iron (II, III) oxide}}{Fe_3O_4} + 4H_2\uparrow$$

15 The diagram shows the reaction between metal and dil. acid.



What is the reason for different behaviour of Mg in test tube B? **Competency Based Que.**

(a) Mg is lighter element than dil. HCl.
(b) Mg reacts with dil. HCl to produce H_2 gas which helps in floating.
(c) Mg reacts with dil. HCl to produce N_2 gas which helps in floating.
(d) Mg reacts with dil. HCl to produce CO_2 gas which helps in floating.

Sol. (*b*) When magnesium reacts with dil.HCl in test tube *B*, then there is formation of salt, i.e. $MgCl_2$ and hydrogen gas comes out as bubbles. These bubbles are responsible for the floating of magnesium metal.

$$\underset{}{Mg} + \underset{\text{(dil.)}}{2HCl} \longrightarrow \underset{\substack{\text{Magnesium}\\\text{chloride}}}{MgCl_2} + H_2 \uparrow$$

16. Metals are refined by using different methods. Which of the following metals is not refined by electrolytic refining?

(a) Au (b) Cu (c) Ca (d) Zn

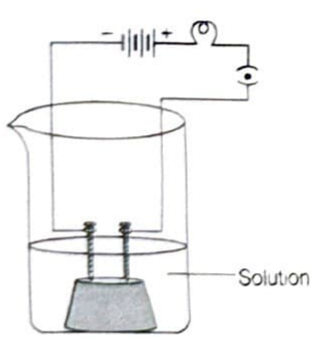
Sol. (*c*) Electrolytic refining is used for metals like Cu, Zn, Ag, Au etc. The method to be used for refining an impure metal depends on the nature of the metal as well as on the nature of impurities present in it.

17. During the electrolytic refining of copper, what happens at the anode? **Competency Based Que.**

(a) Copper ions gain electrons to become neutral copper atoms.
(b) Neutral copper atoms gain electrons to becomes ions.
(c) Copper ions lose electrons to become neutral atoms.
(d) Neutral copper atoms lose electrons to become ions.

Sol. (*d*) During electrolytic refining of copper, the neutral copper atoms lose electrons to become ions.

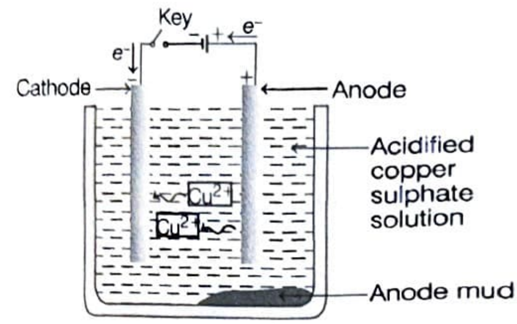
18 In the given experimental set-up, if the experiment is carried out separately with each of the following solutions the cases in which the bulb will glow is **CBSE 2023**



(a) dilute hydrochloric acid
(b) honey
(c) glucose solution
(d) alcohol

Sol. (*a*) HCl dissociates into ions. These ions conduct electricity in the solution resulting in the glowing of bulb. On the other hand, glucose, honey and alcohol do not dissociate into ions. Therefore, bulb will not glow.

19. The following diagram shows the electrolytic refining of copper : **CBSE 2023**



Which of the following statements is incorrect description of the process?

(a) The impure metal from the anode dissolves into the electrolyte.
(b) The pure metal from the electrolyte is deposited on th cathode.
(c) Insoluble impurities settle down at the bottom of the anode.
(d) On passing the current through the electrolyte, the pure metal from the anode dissolves into the electrolyte

Sol. (*a*) Among the given statements regarding electrolytic refining of copper, statement (a) is incorrect.

20. Which among the following statements is incorrect for magnesium metal? **NCERT Exempla**

(a) It burns in oxygen with a dazzling white flame.
(b) It reacts with cold water to form magnesium oxide an evolves hydrogen gas.
(c) It reacts with hot water to form magnesium hydroxide and evolves hydrogen gas.
(d) It reacts with steam to form magnesium hydroxide an evolves hydrogen gas.

Sol. (*b*) Magnesium metal does not react with cold water. Mg reacts with hot water and steam both to give magnesium hydroxide and hydrogen.

$$Mg(s) + H_2O(l) \longrightarrow \underset{\substack{\text{Magnesium}\\\text{hydroxide}}}{Mg(OH)_2} + 2H_2(g) \uparrow$$

21. Which of the following metals, do not corrode in moist air? **CBSE 202**

(a) Copper (b) Iron (c) Gold (d) Silver

Sol. (*c*) Among the given metals gold do not corrode in moist air as it is an unreactive metal.

22. Galvanisation is a method of protecting iron from rusting by coating it with a thin layer of

(a) gallium (b) aluminium (c) zinc (d) silver

Sol. (*c*) Galvanisation is a method of protecting iron from rusting by coating it with a thin layer of zinc (Zn) metal.

23. A piece of zinc (Zn)-a reactive metal-was dropped into a test tube containing a substance. A zinc salt was formed and hydrogen gas was liberated. This is shown in the equation below.

$$Zn^{+} \longrightarrow \text{Zinc salt} + H_2 \text{ gas}$$

Which of the following can be the substance that zinc was dropped into?

P. water

Q. hydrochloric acid

R. a solution of a zinc salt **Competency Based Que.**

(a) Only P (b) Only Q
(c) Only R (d) Either P or R

Sol. (*b*) Zinc is dropped into hydrochloric acid to form a salt (zinc chloride) and liberate hydrogen gas.

$$Zn + 2HCl \longrightarrow ZnCl_2 + H_2 \uparrow$$

24. Bronze is an alloy of **CBSE 2023**

(a) copper and zinc (b) aluminum and tin
(c) copper, tin and zinc (d) copper and tin

Sol. (*d*) Bronze is an alloy of copper and tin only.

25. $2Al + 3H_2O \longrightarrow Al_2O_3 + X$

What is *X* in the reaction?

(a) Al (b) H_2 (c) O_3 (d) AlH_3

Sol. (*b*) Water has no action on aluminium due to layer of oxide on it. When steam is passed over pure heated aluminium, hydrogen is produced.

The complete reaction is as follows

$$2Al + \underset{\text{(Steam)}}{3H_2O} \longrightarrow Al_2O_3 + 3H_2$$

26. Give reasons :

(i) Reactivity of Al decreases if it is dipped in HNO_3.

(ii) Carbon cannot reduce the oxides of Na or Mg.

(iii) NaCl is not a conductor of electricity in solid state whereas it does conduct electricity in aqueous solution as well as in molten state. **CBSE 2019**

(iv) Iron articles are galvanised.

(v) Metals like Na, K, Ca and Mg are never found in their free state in nature. **NCERT Exemplar**

Sol. (i) Reactivity of Al decreases if it is dipped in HNO_3 due to the formation of a coating of aluminium oxide/ Al_2O_3.

(ii) Na or Mg are more reactive metals as compared to carbon. So, their oxides are more stable and carbon can not reduce their oxides.

(iii) In molten state, due to heat the electrostatic forces of attraction between the oppositely charged ions are overcome. So, ions move freely and conduct electricity.

In aqueous solutions ions are free and conduct electricity.

(iv) Iron articles are galvanised to protect them from corrosion.

(v) Reactive metals like Na, K, Ca and Mg react easily with different elements and occur in the form of ores.

27. Listed here is the reactivity of certain metals.

Metal	Reaction with air	Reaction with water	Reaction with dilute acids
Gold	Does not oxidise or burn	No reaction	No reaction
Sodium	Burns vigorously to form oxide	Violent reaction	Violent reaction
Zinc	Burns to form oxides	Reacts on heating with water	Reacts to produce hydrogen
Platinum	Does not oxidise or burn	No reaction	No reaction

Which of the above metals are likely to be obtained in their pure states from the Earth's crust? **CBSE Question Bank**

(a) Only gold
(b) Only sodium
(c) Gold and platinum
(d) Zinc and sodium

Ans. (*c*) As gold and platinum do not react with air, water and dilute acids so they can be obtained in their pure states from the Earth's crust.

Assertion-Reason Questions

Direction (Q. Nos. 1-9) ***In each of the following questions, a statement of Assertion is given by the corresponding statement of Reason. Of the statements, mark the correct answer as.***

(a) Both Assertion and Reason are true and Reason is the correct explanation of Assertion.
(b) Both Assertion and Reason are true, but Reason is not the correct explanation of Assertion.
(c) Assertion is true, but Reason is false.
(d) Assertion is false, but Reason is true.

1. Assertion (A) Oxides of metals show basic characters.

Reason (R) Oxides of metals react with acid to form salt and water.

Sol. (*a*) Both Assertion and Reason are true and Reason is the correct explanation of Assertion.

2. Assertion (A) In the following reaction.

$$ZnO + C \longrightarrow Zn + CO$$

ZnO undergoes reduction.

Reason (R) Carbon is a reducing agent that reduces ZnO to Zn. **CBSE 2023**

Sol. (*a*) Both Assertion and Reason are true and Reason is the correct explanation of Assertion.

3. **Assertion**(A)Food cans are coated with tin and not with zinc.

Reason (R) Zinc is more reactive than tin.

Sol. (*a*) Both Assertion and Reason are true and Reason is the correct explanation of Assertion.

4. **Assertion** (A) Carbon reacts with oxygen to form carbon dioxide which is an acidic oxide.

Reason (R) Non-metals form acidic oxides.

Sol. (*a*) Both Assertion and Reason are true and Reason is the correct explanation of Assertion.

5. **Assertion** (A) Alloys are commonly used in electrical heating devices like electric iron and heater.

Reason (R) Alloys have lower melting points than their constituent metals.

Sol. (*c*) Assertion is true but Reason is false.

The correct reason is that, generally the melting point of alloys is higher than their constituent metals, so alloys is commonly used in electrical heating devices like electric iron and heater.

6. **Assertion** (A) A piece of zinc metal gets reddish brown coating when kept in copper sulphate solution for some time. **CBSE 2024**

Reason (R) Copper is more reactive metal than zinc.

Sol. (*c*) Assertion is true but Reason is false. The correct form of R is : Zinc is more reactive in comparison of copper and can easily displace copper from copper sulphate solution.

7. **Assertion** (A) Zinc carbonate is heated strongly in presence of air to form zinc oxide and carbon dioxide.

Reason (R) Calcination is the process in which a carbonate ore is heated strongly in the absence of air to convert into metal oxide.

Sol. (*d*) Assertion is false but Reason is true. The corrected form of Assertion is :

Zinc carbonate is heated in the absence of air to form zinc oxide and carbon dioxide.

8. **Assertion** (A) Magnesium chloride is an ionic compound.

Reason (R) Metals and non-metals react by mutual transfer of electrons.

Sol. (*a*) Both Assertion and Reason are true and Reason is the correct explanation of Assertion.

9. **Assertion** (A) Highly reactive metals are obtained by electrolytic reduction.

Reason (R) In the electrolytic reduction, metal is deposited at the cathode.

Sol. (*b*) Both Assertion and Reason are true but Reason is not the correct explanation of Assertion.

Highly reactive metals are obtained by electrolytic reduction because these metals have more affinity for oxygen than carbon so they cannot be obtained by reduction with carbon.

Case Based/Sources Based Questio

Direction (Q. Nos. 1-6) ***Answer the questions on basis of your understanding of the following passage and related studied concepts:***

1. **The reactivity series is a list of metals arrange the order of their decreasing activities. Copper silver, both being placed lower than hydrogen in reactivity series are easily displaced out solution of their ions by reactive metals higher the reactivity series. Iron, however cannot disp Na^+ and Ca^{2+} as it present below Na and Ca in reactivity series.**

(i) Name two metals which will displace hydrogen from dilute acids.

Sol. Zinc and magnesium

(ii) 5 mL each of concentrated HCl, HNO_3 and mixture of concentrated HCl (15 mL) and concentrated HNO_3 (5 mL) were taken in a test tube labelled as *A*, *B* and *C*. *A* small pie of metal was put in each test tube. No cha occurred in test tube *A* and *B* but the meta dissolved in the tube *C*. Identify the metal present in test tube *C*.

Sol. The metal is gold which got dissolved in *aqua-regia*.

(iii) Zinc is used in the galvanisation of iron and not copper.

Sol. Zinc is placed above iron in the activity series. When deposited over its surface (galvanisation), zinc gets corroded and not iron. But copper is placed below iron and therefore, does not participate in the corrosion. In this case, iron is corroded or rusted.

Or (iii) Name the metals that can be displaced, wh excess of iron fillings is added to a solution containing a mixture of ions Pb^{2+}, Zn^{2+}, Hg^{2} and Mg^{2+}.

Sol. Iron can displace lead (Pb) and mercury (Hg) fro their solutions as iron is more reactive than lead and mercury. This is because it is placed above them in the reactivity series of the metals.

2. **The reactivity series is a list of metals arranged the order of their decreasing activities.**

On the basis of their reactive tendency to lo electron and their reactive nature, metals a arranged in a series, this series is called activi series or reactivity series of metals. The meta that are placed above hydrogen are called mo reactive metals and the metals that are place below hydrogen are called least reactive metals The most reactive metals can displaces les reactive metals from its salts solution.

K	Potassium	Most reactive
Na	Sodium	
Ca	Calcium	
Mg	Magnesium	
Al	Aluminium	
Zn	Zinc	Reactivity decrease ↓
Fe	Iron	
Pb	Lead	
H	Hydrogen	
Cu	Copper	
Hg	Mercury	
Ag	Silver	
Au	Gold	Least reactive

(i) Which metals exist in their native states in nature and why?

Sol. Au and Ag are placed at the end of the reactivity series and the metals which come below hydrogen are least reactive. Hence, they exist in their native states in nature. [1]

(ii) Which method is used to extract the metals present at the top of the series?

Sol. Metals which are placed at the top of the reactivity series have high reactivity towards oxygen. Hence, the method of electrolytic reduction is employed for metals, like Na, Mg, Ca etc. [1]

(iii) What change would you observe when a copper wire is dipped in an aqueous solution of silver nitrate for sometime?

Sol. It is a chemical reaction where a more reactive element i.e. copper displaces the less reactive one i.e. silver from its salt solution. The colour of the solution changes from colourless to blue and a silver solid deposits on copper wire. [2]

Or

(iii) What happens when calcium react with nitric acid? Also name any two metals which react with steam but not with hot water.

Sol. Calcium reacts with nitric acid to form calcium nitrate, dinitrogen monoxide and water. Metals like aluminium and zinc do not react with hot or cold water. They react only with steam to form a metal oxide and hydrogen. [2]

3. The melting points and boiling points of some ionic compounds are given below: **CBSE 2023**

Compound	Melting point (K)	Boiling point (K)
NaCl	1074	1686
LiCl	887	1600
$CaCl_2$	1045	1900
CaP	2850	3120
$MgCl_2$	981	1685

These compounds are termed ionic because they are formed by the transfer of electrons from a metal to a non-metal. The electron transfer in such compounds is controlled by the electronic configuration of the elements involved. Every element tends to attain a completely filled valence shell of its nearest noble gas or a stable octet.

(i) Show the electron transfer in the formation of magnesium chloride.

Sol. Refer to text on Pg-64 (Reaction between metals and non-metals). [1]

(ii) List two properties of ionic compounds other than their high melting and boiling points.

Sol. Refer to text on Pg-65 (Properties of ionic compounds). [1]

(iii) While forming an ionic compound say sodium chloride how does sodium atom attain its stable configuration?

Sol. Refer to text on Pg-64 (Reaction between metals and non-metals). [2]

Or

(I) Why do ionic compounds in the solid state not conduct electricity?

(II) What happens at the cathode when electricity is passed through an aqueous solution of sodium chloride?

Sol. (i) Refer to text on Pg- 65 (Conduction of electricity). [1]

(ii) Refer to text on Pg-67 (Electrolytic reduction). [1]

4. Two students decided to investigate the effect of water and air on iron object under identical experimental conditions. They measured the mass of each object before placing it partially immersed in 10 mL of water. After a few days, the object were removed, dried and their masses were measured. The table shows their results. **CBSE SQP 2022-23**

Student	Object	Mass of object before rusting in g	Mass of the coated object in g
A	Nail	3.0	3.15
B	Thin plate	6.0	6.33

(i) What might be the reason for the varied observations of the two students?

Sol. The main reason for varied observations of the two students is rusting that occured in both A and B. So, there is an increase in mass in both the cases. However, the surface area of B is more thus, extent of rusting is more. [1]

(ii) In another set up the students coated iron nails with zinc metal and noted that, iron nails coated with zinc prevents rusting. They also observed that zinc initially acts as a physical barrier, but an extra advantage of using zinc is that it continues to prevent rusting even, if the layer of zinc is damaged. Name this process of rust prevention and give any two other methods to prevent rusting.

Sol. This process is named as galvanisation which is used for rust prevention. Oiling and greasing are other two methods used to prevent rusting. [1]

(iii) In which of the following applications of Iron, rusting will occur most? Support your answer with valid reason.

A. Iron bucket electroplated with zinc.

B. Electricity cables having iron wires covered with aluminium.

C. Iron hinges on a gate.

D. Painted iron fence.

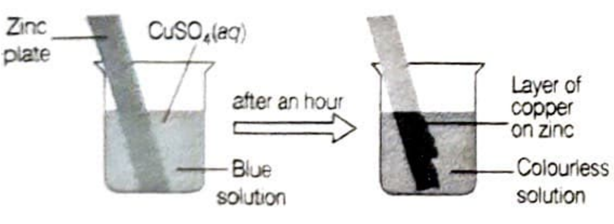
Sol. The rusting will occur most is in iron hinges which are present on a gate. This is due to the fact that iron is in direct contact with both atmospheric oxygen and moisture or water vapours. In rest of the applications, rusting will be occur at very slow rate. [2]

Or

(iii) How is pure iron used for making different alloys?

Sol. Alloying is a process of mixing of other metals to a particular metal so that its properties can be improved. Iron is soft in its pure state. It can be mixed easily with other element to make desired substance. e.g. Steel is an alloy of iron and carbon. [2]

5. Reena immersed a zinc plate in an aqueous solution of copper sulphate. She noticed a thick layer of copper on the surface of the zinc plate after an hour.



(i) What should Reena have done to make the reaction faster?

Sol. Reena should have choosed zinc flakes instead of zinc plate.The reaction with copper sulphate takes place faster with small zinc flakes than thicker zinc plate.

$$Zn(s) + CuSO_4(aq) \longrightarrow ZnSO_4(aq) + Cu(s)$$ [1]

(ii) $2Al + 3H_2O \longrightarrow Al_2O_3 + X$

What is X in the reaction?

Sol. The reaction is as follows

$$2Al + 3H_2O \longrightarrow Al_2O_3 + 3H_2 \uparrow$$

Therefore, 'X' is hydrogen gas.

(iii) No reaction takes place when a copper plate immersed in an aqueous solution of zinc sulp Explain the reason behind this.

Sol. No, reaction will take place when the copper metal dipped in the solution of zinc sulphate because zinc more reactive than copper, it won't be able to displa zinc from zinc sulphate.

Or

(iii) What makes gold exist in free state in nature?

Sol. Gold is present at the bottom of the activity serie it is least reactive. It does not combine with other elements and generally found in free state only.

6. Metals are required for a variety of purposes. Fo we need their extraction from their ores. Ores m from the earth are usually contaminated with impurities which must be removed prior to extraction of metals. The extraction of pure m involves the following steps: **CBSE**

(a) Concentration of ore

(b) Extraction of the metal from the concentrated

(c) Refining of the metal

(i) Name an ore of mercury and state the form in which mercury is present in it.

Sol. Refer to text on Pg-66 ([Extraction of metals of low reactivity].

(ii) What happens to zinc carbonate when it is he strongly in a limited supply of air?

Sol. Refer to text on Pg-66-67 [Extraction of metals of medium reactivity].

(iii) The reaction of a metal A with Fe_2O_3 is highly exothermic and is used to join railway tracks.

(1) Identify the metal A and name the reaction taking place.

(2) Write the chemical equation for the reactio metal A with Fe_2O_3.

Sol. (1) and (2) Refer to Pg-67 (Thermit welding).

Or (iii) We cannot use carbon to obtain sodium from sodium oxide. Why? State the reactions taking place at cathode and anode during electrolytic reduction of sodium chloride.

Sol. Refer to text on Pg-67 (Extraction of metals of reactivity).

Constructed Response (Descriptive) Questions

Very Short Answer Type Questions

1. (i) Which of the following metals will melt at body temperature (37°C)?
Gallium, magnesium, caesium and aluminium.

(ii) Name two metals which react with dil. HNO_3 to evolve hydrogen gas.

Sol. (i) Among the given metals gallium and caesium melt at body temperature (37°C). [1]

(ii) Manganese (Mn) and magnesium (Mg) are the two metals that react with dil. HNO_3 to evolve hydrogen gas. [1]

2. (i) An element forms an oxide A_2O_3 which is acidic in nature. Identify A as a metal or non-metal. **NCERT Exemplar**

(ii) A solution of $CuSO_4$ was kept in an iron pot. After few days the iron pot was found to have a number of holes in it. Explain the reason in terms of reactivity. Write the equation of the reaction involved.

Sol. (i) Oxides of non-metals are acidic in nature while those of metals are basic. As the element A forms an acidic oxide, therefore, A must be a non-metal. Also it should have + 3 charge, i.e. it should have 3 valence electrons. Therefore, A is boron,(electronic configuration is 2, 3) and its oxide is B_2O_3. [1]

(ii) As iron is more reactive metal than copper, it displaces copper from copper sulphate solution. The reaction takes place as follows

$$\underset{\text{Iron}}{Fe(s)} + \underset{\text{Copper (II) sulphate}}{CuSO_4(aq)} \longrightarrow \underset{\text{Iron (II) sulphate}}{FeSO_4(aq)} + \underset{\text{Copper}}{Cu(s)}$$

Since, iron takes part in this reaction, therefore holes are produced at places where iron metal has reacted to form iron (II) sulphate. [1]

3. (i) Arrange the following metals in the decreasing order of reactivity Na, K, Cu and Ag.

(ii) Although, metals form basic oxides, name one metal which forms an amphoteric oxide. **Competency Based Que.**

Sol. (i) The decreasing order of reactivity of the given metals is

$$K > Na > Cu > Ag. \quad [1]$$

(ii) Aluminium is a metal which forms an amphoteric oxide. [1]

4. A metal M does not liberate hydrogen from acids but reacts with oxygen to give a black colour product. Identify M and black coloured product and also explain the reaction of M with oxygen. **NCERT Exemplar**

Sol. Metal M, Cu as it is less reactive than hydrogen. So, it does not react with acid to liberate hydrogen.

$$\underset{(M)}{Cu(s)} + \underset{\text{Dil. hydrochloric acid}}{HCl} \longrightarrow \text{No reaction} \quad [1]$$

Copper reacts with the oxygen of air on prolonged heating to form a black substance, copper (II) oxide.

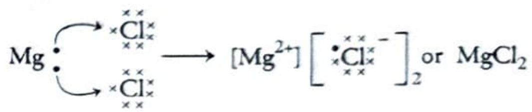
$$\underset{\text{Copper}}{2Cu(s)} + O_2(g) \longrightarrow \underset{\text{Copper (II) oxide (black)}}{2CuO(s)} \quad [1]$$

5. (i) Name one metal which reacts neither with cold water, nor with hot water, but reacts with steam to produce hydrogen gas. **NCERT Exemplar**

(ii) Show the electron transfer in the formation of $MgCl_2$ from its elements.

Sol. (i) Iron is the metal which does not react with cold and hot water but reacts with steam to produce hydrogen gas. [1]

(ii) The electron transfer in the formation of $MgCl_2$ from its elements is shown below.



[1]

6. (i) A piece of granulated zinc was dropped into copper sulphate solution. After sometime, the colour of the solution changed from blue to colourless. Why?

(ii) Arrange the following metals in decreasing order of their reactivity. **Competency Based Que.**

Fe, Zn, Na, Cu, Ag

Sol. (i) Blue copper sulphate is converted into colourless zinc sulphate, as zinc, being more reactive, displaces copper from $CuSO_4$ solution and forms a colourless solution of zinc sulphate. [1]

$$Zn(s) + \underset{\text{(Blue)}}{CuSO_4(aq)} \longrightarrow \underset{\text{(Colourless)}}{ZnSO_4(aq)} + Cu(s)$$

(ii) The correct decreasing order of reactivity of the metals.

$$Na > Zn > Fe > Cu > Ag \quad [1]$$

7. (i) A cleaned aluminium foil was placed in an aqueous solution of zinc sulphate. When the aluminium foil was takenout of the zinc sulphate solution after 15 minutes, its surface was found to be coated with a silvery grey deposit. From the given observation, what can be concluded?

(ii) Why does aluminium not react with water under ordinary condition?

Sol. (i) Aluminium being more reactive than zinc displaces zinc from zinc sulphate solution. [1]

$$\underset{\text{Zinc sulphate}}{3ZnSO_4(aq)} + \underset{\text{Aluminium foil}}{2Al(s)} \longrightarrow \underset{\text{Aluminium sulphate}}{Al_2(SO_4)_3(aq)} + \underset{\text{Zinc}}{3\,Zn(s)}$$

(ii) Aluminium does not react with water under ordinary condition because of the presence of a thin layer of aluminium oxide on its surface.

8. Aluminium is a reactive metal but is still used for packing food articles. Give reason.

Sol. Aluminium is strong and cheap metal. It is also a good conductor of heat. But it is highly reactive. When it is exposed to moist air, its surface is covered with thin impervious layer of aluminium oxide (Al_2O_3). This layer does not allow moist air to come in contact with fresh metal and hence, protects the metal underneath from further damage or corrosion. [1]

Thus, after the formation of this protective layer of Al_2O_3, aluminium becomes resistant to corrosion. It is because of this reason that although aluminium is a lightly reactive metal, it is still used in food packaging. [1]

9. Give the reaction involved during extraction of zinc from its ore by

(i) roasting of zinc ore.

(ii) calcination of zinc ore.

Sol. (i) **Roasting of zinc ore** Roasting is done for the sulphide ore. Sulphide ore of zinc is ZnS (zinc blende).

$$\underset{\substack{\text{Zinc sulphide}\\ \text{(zinc blende ore)}}}{2ZnS(s)} + \underset{\substack{\text{Oxygen}\\ \text{(from air)}}}{3O_2} \xrightarrow{\text{Roasting}} \underset{\text{Zinc oxide}}{2ZnO(s)} + 2SO_2(g)$$

ZnO is then reduced to zinc by heating with coke (carbon).

$$\underset{\text{Zinc oxide}}{ZnO(s)} + \underset{\substack{\text{Carbon}\\ \text{(reducing agent)}}}{C(s)} \xrightarrow{\Delta} \underset{\text{Zinc metal}}{Zn(s)} + \underset{\text{Carbon monoxide}}{CO(g)}$$ [1]

(ii) **Calcination of zinc ore** Calcination is done for the carbonate ore. Carbonate ore of Zn is calamine ($ZnCO_3$).

$$\underset{\text{Zinc carbonate}}{ZnCO_3(s)} \xrightarrow{\text{Calcination}} \underset{\text{Zinc oxide}}{ZnO(s)} + \underset{\text{Carbon dioxide}}{CO_2(g)\uparrow}$$

ZnO is then reduced to zinc by heating with coke (carbon).

$$\underset{\text{Zinc oxide}}{ZnO(s)} + \underset{\text{Carbon}}{C(s)} \xrightarrow{\Delta} \underset{\text{Zinc metal}}{Zn(s)} + \underset{\text{Carbon monoxide}}{CO(g)}$$ [1]

10. Show the formation of MgO by the transfer of electrons in the two elements using electron dot structures.

Sol.

$$\underset{(2,\,8,\,2)}{Mg} \longrightarrow \underset{\substack{(2,\,8)\\ \text{(Magnesium cation)}}}{Mg^{2+}} + 2e^-$$ [½]

$$\underset{(2,\,8,\,6)}{O} + 2e^- \longrightarrow \underset{(2,\,8)}{O^{2-}}$$ [½]

$$Mg\colon + \overset{\times\times}{\underset{\times\times}{O}} \longrightarrow [Mg^{2+}][O^{2-}]$$

$$Mg^{2+} + O^{2-} \longrightarrow MgO$$ [1]

11. A metal X combines with a non-metals Y and transfer electrons to form a compound Z.

(i) State the type of bond in compound Z.

(ii) What can you say about the melting point and boiling point of compound Z?

Sol. X being a metal loses electrons Y being a non-metal g[ains] electrons from X to form Z.

(i) The chemical bond formed by the transfer of electron[s] from one atom to another is known as an ionic bond. Hence, Z is an ionic compound.

(ii) Compound Z is an ionic compound thus, it has high melting and boiling points.

12. (i) Define ionic compounds. Name one proper[ty] which is not shown by ionic compounds.

(ii) A green layer is gradually formed on a copper pla[te] when left exposed to air for a week in a bathroom. What could this green substance be?

Sol. (i) The compounds formed by the complete transfer of electrons from one atom to another is called ionic compounds.

Ionic compounds do not conduct electricity in the solid state.

(ii) This green substance is basic copper carbonate $CuCO_3 \cdot Cu(OH)_2$.

13. (i) Metals are refined by using different method[s]. Which of the following metals are refined by electrolytic refining?

Au, Cu, Na and K

(ii) Name an alloy of lead and thin.

Sol. (i) Electrolytic refining is used for metals like Cu, Zn, Ag, Au etc.

This method is used for refining an impure metal that depends on the nature of the metal as well as on the nature of impurities present in it. Cu and Au are refined by electrolytic refining technique.

(ii) Solder is an alloy of lead and tin.

14. A metal that exists as a liquid at room temperature is obtained by heating its sulphide in the presence of a[ir]. Identify the metal and its ore and give the reaction involved. **CBSE 2019 [NCERT Exemplar]**

Sol. Metals low in activity series can be obtained by reducing their sulphides or oxides by heating. Mercury is the only metal that exists as liquid at room temperature. It can be obtained by heating cinnabar (HgS), the sulphide ore of mercury.

The reaction are as follows

$$2HgS(s) + 3O_2(g) \xrightarrow{\text{Heat}} 2HgO(s) + 2SO_2(g)$$

$$2HgO(s) \xrightarrow{\text{Heat}} 2Hg(l) + O_2(g)$$

15. (i) Name two metals that are obtained by electrolys[is] of their chlorides in molten form.

(ii) Name an alloy that contains a non-metal as one of its constituents. **NCERT Exemplar**

Sol. (i) Sodium and calcium are obtained by electrolysis of their chlorides in molten form. [1]

(ii) Steel (iron + carbon) is an alloy that contains a non-metal as one of its constituents. [1]

16. (i) Name an alloy which has mercury as one of its constituents.

(ii) How is Zn refined?

Sol. (i) Zinc amalgam is an alloy that has mercury as one of its constituents. [1]

(ii) Zinc (Zn) is refined by using electrolytic refining method. In this process, a thick block of impure metal is used as anode and thin strip of pure metal is used as cathode. [1]

17. A metal that exists as a liquid at room temperature is obtained by heating its sulphide in the presence of air. Identify the metal and its ore and give the reactions involved. NCERT Exemplar

Sol. The metal is mercury (Hg) because it is the only metal which exists in liquid state at room temperature. Mercury is obtained from the sulphide ore called cinnabar ore (HgS). The reactions involved are

$$\underset{\text{Mercury (II) sulphide (cinnabar ore)}}{2HgS(s)} + \underset{\text{Oxygen (from air)}}{3O_2(g)} \xrightarrow{\text{Roasting}} \underset{\text{Mercury (II) oxide}}{2HgO(s)} + \underset{\text{Sulphur dioxide}}{2SO_2(g)}$$ [1]

$$\underset{\text{Mercury (II) oxide (formed above)}}{2HgO(s)} \xrightarrow[\text{reduction}]{\text{Heat}} \underset{\text{Mercury metal}}{2Hg(l)} + O_2(g)$$ [1]

18. Explain what happens if bauxite containing iron and silica as impurities is directly subjected to the process of electrolytic reduction without prior purification.

Sol. Bauxite, the most important ore of aluminium contains only 30-70% aluminium oxide (alumina).

Crude bauxite ($Al_2O_3 \cdot 2H_2O$) contains iron oxide (Fe_2O_3) and silica (SiO_2) as impurities.

If any amount of iron is present in the bauxite, it will get deposited at the cathode in preference to aluminium because iron is less electropositive than aluminium. [2]

19. (i) A metal oxide on being heated with carbon does not produce carbon dioxide. Give a possible explanation for this behaviour of the metal oxide.

(ii) A metallic element M, has the following properties.

- floats on water

- can be cut with a knife

- occurs naturally as its chloride, of formula, MCl_2

- its oxide dissolves in water to form the hydroxide.

(a) State the method of manufacture of the metal M.

(b) Name the major byproduct obtained in the process. Competency Based Que.

Sol. (i) The highly reactive metals such as K, Na, Ca and Al are placed up in the reactivity series. The oxides of highly reactive metals are very stable and cannot be reduced by carbon. [1]

(ii) The metallic element described in the question is likely to be magnesium.

This is because magnesium can float on water, can be cut with a knife its oxide dissolves in water to form the hydroxide and it occurs naturally as magnesium chloride ($MgCl_2$).

(a) The method of manufacturing is primarily through electrolysis of its molten chloride. [1/2]

(b) The major byproduct obtained in this manufacturing process is chlorine gas, produced during the electrolysis of magnesium chloride. [1/2]

20. A metal A, which is used in thermit process, when heated with oxygen gives an oxide B which is amphoteric in nature. Identify A and B. Write down the reactions of oxide B with HCl and NaOH. NCERT Exemplar

Sol. Metal A is aluminium (Al) which is used in thermit process. Al reacts with oxygen to form aluminium oxide Al_2O_3 (B) which is amphoteric in nature.

$$4Al + 3O_2 \longrightarrow \underset{(B)}{2Al_2O_3(s)}$$

(i) $$\underset{\text{Aluminium oxide}}{Al_2O_3(s)} + \underset{\text{Hydrochloric acid}}{6HCl(aq)} \longrightarrow \underset{\text{Aluminium chloride}}{2AlCl_3(aq)} + 3H_2O(l)$$ [1]

(ii) $$\underset{\text{Aluminium oxide}}{Al_2O_3} + \underset{\text{Sodium hydroxide}}{2NaOH(aq)} \longrightarrow \underset{\text{Sodium aluminate}}{2NaAlO_2(aq)} + H_2O(l)$$ [1]

21. When a metal X is treated with cold water, it gives a basic salt Y with molecular formula XOH (molecular mass = 40) and liberates a gas Z which easily catches fire. Identify X, Y and Z and also write the reaction involved. NCERT Exemplar

Sol. Molecular formula of $Y = XOH$

Let the atomic weight of metal X is a.

Then, molecular mass of $XOH = a + 16 + 1 = 40$ (Given)

$$a = 40 - 17 = 23$$

Thus, metal X is sodium (Na) because 23 is the atomic weight of sodium metal. Sodium reacts with water as [1]

$$2Na + \underset{\text{(Cold)}}{2H_2O} \longrightarrow \underset{Y}{2NaOH} + H_2\uparrow$$

Sodium (Na) gives hydrogen gas (Z) on reaction with cold water.

So, Y is NaOH (sodium hydroxide).

Z is H_2 (hydrogen gas). [1]

Short Answer Type Questions

1. A reddish brown metal used in electrical wires when powdered and heated strongly turns black. When hydrogen gas is passed over this black substance, it regains its original colour. Based on this information answer the following questions. **CBSE 2023**

(i) Name the metal and the black substance formed.

(ii) Write balanced chemical equations for the two reactions involved in the above information.

Sol. Refer to text on Pg 61-62 (Reaction of metals with oxygen) [3]

2. State reasons for the following. **Competency Based Que.**

(i) Zinc oxide is an amphoteric oxide.

(ii) Sodium metal is stored in bottle filled with metals, generally hydrogen gas is not evolved.

(iii) In the reactions of nitric acid with metals generally hydrogen gas is not evolved. **CBSE 2024**

Sol. (i) Zinc oxide reacts with both acids and alkalis. Therefore, it is an amphoteric oxide.

$$ZnO + \underset{(acid)}{2HCl} \longrightarrow ZnCl_2 + H_2O$$

$$ZnO + \underset{(base)}{2NaOH} \longrightarrow Na_2ZnO_2 + H_2O \quad [1]$$

(ii) Sodium is stored in kerosene because sodium reacts vigorously with oxygen and moisture due to its high reactivity to keeping it in kerosene will prevent sodium from coming in contact with oxygen and moisture. [1]

(iii) Hydrogen gas is not evolved when a metal reacts with nitric acid because nitric acid is a strong oxidising agent, it oxidises the hydrogen to produce water. [1]

3. (i) By the transfer of electrons, illustrate the formation of bond in magnesium chloride and identify the ions present in this compound.

(ii) Ionic compounds are solids. Give reason.

(iii) With the help of a labelled diagram show the experimental set up of action of steam on a metal. **CBSE (All India) 2020**

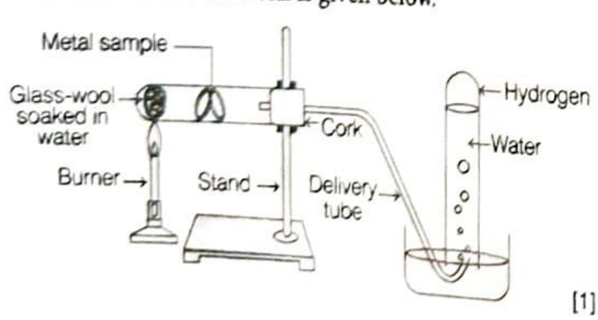
Sol. (i) The formation of bond in magnesium chloride is as follows

Refer to text on Page-64 (Ionic bond formation).

The ions present in this compound are Mg^{2+} and Cl^-. [1]

(ii) Refer to text on Pg-65 (Properties of ionic compounds). [1]

(iii) The diagram that shows the experimental set up of action of steam on metal is given below.



[1]

4. An alkali metal A gives a compound B (molecular mass = 40) on reacting with water. The compound B gives soluble compound C on treatment with aluminium oxide. Identify A, B and C and give the reaction involved. **NCERT Exemplar**

Sol. Let, the atomic weight of an alkali metal A is x. When it reacts with water it forms a compound B having molecular mass 40. Let, the reaction is

$$2A + 2H_2O \longrightarrow \underset{(B)}{2AOH} + H_2 \uparrow$$

According to question, molecular mass of compound

$$B = x + 16 + 1 = 40$$

$$\therefore \quad x = 40 - 17 = 23 \quad \text{(Given)}$$

It is the atomic weight of Na (sodium), therefore, the alkali metal (A) is Na and the reaction is

$$\underset{(A)}{2Na} + 2H_2O \longrightarrow \underset{(B)}{2NaOH}(aq) + H_2(g)\uparrow$$

So, compound B is sodium hydroxide (NaOH) which reacts with aluminium oxide (Al_2O_3) to give sodium aluminate ($NaAlO_2$). Thus, C is sodium aluminate, $NaAlO_2$.

The reaction involved is

$$Al_2O_3 + \underset{(B)}{2NaOH} \longrightarrow \underset{\substack{\text{Sodium aluminate} \\ (C)}}{2NaAlO_2} + H_2O$$

5. State giving reason the reduction process to obtain the following metals from their compounds: **CBSE 2024**

(i) Mercury **(ii) Copper and**

(iii) Sodium

Sol. (i) Mercury (Hg) is a less reactive metal. It is obtained simply by heating their oxides.

$$2HgO \xrightarrow{\Delta} \underset{\substack{\text{(Mercury} \\ \text{metal)}}}{2Hg} + O_2 \text{ (Reduction)}$$

(ii) Copper (Cu) is also a less reactive metal. It is found in sulphide ore (Copper glance) which on oxidation form copper oxide then on self reduction or auto-reduction process give Cu metal.

$$2Cu_2S + 3O_2 \longrightarrow 2Cu_2O + 2SO_2$$

$$Cu_2S + 2Cu_2O \longrightarrow \underset{\substack{\text{(Copper} \\ \text{metal)}}}{6Cu} + SO_2$$

(Self reduction)

(iii) Sodium (Na) is highly reactive metal, so it obtained by electrolytic reduction of molten ores, e.g. in NaCl.

$$NaCl \xrightarrow{\Delta} Na^+ + Cl^-$$

At cathode, $Na^+ + e^- \longrightarrow Na$

At anode $\quad 2Cl^- \longrightarrow Cl_2 + 2e^-$

6. State the property utilised in the following:

(i) Graphite in making electrodes.

(ii) Electrical wires are coated with polyvinyl chloride (PVC) or a rubber-like material.

(iii) Metal alloys are used for making bells and strings of musical instruments.

Sol. (i) Graphite is an allotrope of carbon which is a good conductor of electricity. It is cheap, insoluble in water, do not react with acids and bases and is non-corrosive material. Due to these properties, it is used in making electrodes. [1]

(ii) Polyvinyl chloride (PVC) or a rubber-like material are insulators and hence do not allow the flow of current and prevent from current shock. Hence, these are used in coating the electrical wires. [1]

(iii) Metals and metal alloys are generally sonorous in nature, i.e. they produce sound. Due to this property, they are used for making bells and strings of musical instruments. [1]

7. A student was given Mg, Zn, Fe and Cu metals. He puts each of them in dil.HCl contained in different test tubes. Identify which of them

(i) will not displace H_2 from dil. HCl?

(ii) will give H_2 with 5% HNO_3?

(iii) will be displaced from its salt solution by all other metals?

Sol. (i) Cu metal will not displace H_2 from dil. HCl. [1]

(ii) Mg will give H_2 with 5% HNO_3. [1]

(iii) Cu metal will be displaced from its salt solution by all other metals. [1]

8. (i) Write any two properties of ionic compounds.

(ii) Show the formation of aluminium chloride by the transfer of electrons between the atoms. (Atomic number of aluminium and chlorine are 13 and 17 respectively).

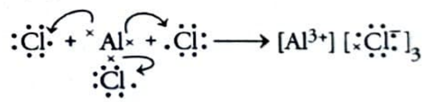
Sol. (i) Properties of ionic compounds are

(a) They are hard crystalline solids. [1]

(b) These compounds have high melting and boiling points as large amount of energy is required to break strong electrostatic forces of attraction. [1]

(ii) $Al = 2, 8, 3$ $\quad Al \longrightarrow Al^{3+} + 3e^-$,

$Cl = 2, 8, 7$ $\quad 3Cl + 3e^- \longrightarrow 3Cl^-$



[1]

9. *P*, *Q* and *R* are 3 elements which undergo chemical reactions according to the following equations:

(a) $P_2O_3 + 2Q \longrightarrow Q_2O_3 + 2P$

(b) $3RSO_4 + 2Q \longrightarrow Q_2(SO_4)_3 + 3R$

(c) $3RO + 2P \longrightarrow P_2O_3 + 3R$

Answer the following questions:

(i) Which element is most reactive?

(ii) Which element is least reactive?

(iii) State and define the type of reaction listed above.

Sol. (i) Most reactive metal is *Q* as it has replaced both *P* and *R* from their compounds. [1]

(ii) Element *R* is least reactive as it has been replaced by both *P* and *Q*. [1]

(iii) The type of reaction listed above is displacement reaction in which a more reactive element displaces a less reactive element from a compound. [1]

10. State giving reason for the change in appearance observed when each of the following metal is exposed to atmospheric air for some time: **CBSE 2024**

(i) Silver, **(ii) Copper and**

(iii) Iron

Sol. (i) When silver is exposed to air for sometime, it acquires a black colour appearance due to the formation of thin layer of sulphide on its surface. The reaction involved is

$$2Ag + H_2S \longrightarrow \underset{\text{(Black)}}{Ag_2S} + H_2 \quad [1]$$

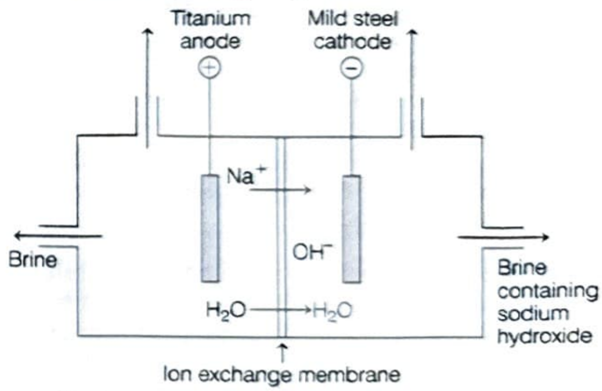
(ii) When copper is exposed to atmospheric air, it reacts with water, CO_2 and oxygen to form basic copper carbonate called malachite which is green in colour. The reaction involved is

$$Cu + H_2O + CO_2 + O_2 \longrightarrow \underset{\substack{\text{(Basic copper carbonate}\\ \text{Malachite green)}}}{CuCO_3 \cdot Cu(OH)_2} \quad [1]$$

(iii) Iron on reaction with the atmospheric oxygen and moisture undergoes corrosion form its oxide called rust which is reddish brown in colour. The reaction involved is

$$2Fe + xH_2O + 3O_2 \longrightarrow \underset{\text{Rust (Reddish brown)}}{Fe_2O_3 . xH_2O} \quad [1]$$

11. Consider the following diagram.



(i) Identify the gases evolved at the anode and cathode in the above experimental set up.

(ii) Name the process that occurs. Why is it called so?

(iii) Illustrate the reaction of the process with the help of a chemical equation. **Competency Based Que.**

Sol. (i) Chlorine gas evolved at anode while hydrogen gas evolved at cathode. [1]

(ii) The name of the process is chlor alkali process. Since, the product obtained are alkali, chlorine gas and hydrogen gas, the process is called chlor alkali process. [1]

(iii) The reaction involved in this process is

$$\underset{\text{Sodium chloride}}{2NaCl(aq)} + \underset{\text{Water}}{2H_2O(l)} \longrightarrow \underset{\text{Sodium hydroxide}}{2NaOH(aq)} + \underset{\text{Chlorine gas}}{Cl_2(g)} + \underset{\text{Hydrogen gas}}{H_2(g)}$$ [1]

12. With the help of suitable chemical equations, list the two main difference between roasting and calcination. How is metal reduced from the product obtained after roasting/calcination of the ore? Write the chemical equation for the reaction involved. **CBSE 2023**

Sol. Refer to text on Pg-67 (Roasting, calcination).

13. A metal acts as a good reducing agent. It reduces Fe_2O_3 and MnO_2. The reaction with Fe_2O_3 is used for joining broken railway tracks. Identify the metal and write all the chemical reactions.

Sol. Metal which reduces Fe_2O_3 and MnO_2 is aluminium, Al. [1]

$$3MnO_2(s) + 4Al(s) \longrightarrow 3Mn(l) + 2Al_2O_3(s) + \text{Heat}$$ [1]

$$Fe_2O_3(s) + 2Al(s) \longrightarrow 2Fe(l) + Al_2O_3(s) + \text{Heat}$$ [1]

14. Account for the following:
(i) Electrical wires are coated with plastic.
(ii) Carbon is not used for reducing aluminium from aluminium oxide.
(iii) Why are cooking utensils made of metal?

Sol. (i) Electrical wires are made-up of copper and aluminium. These wires are coated with plastic to make it safe and shock free. [1]

(ii) Because aluminium is more reactive than carbon therefore, carbon cannot reduce alumina (Al_2O_3) to aluminium. [1]

(iii) Because metals are good conductors of heat. It evenly distributes the heat to the utensils and so, food is cooked. [1]

15. How is the method of extraction of metals high up in the reactivity series different from that for metals in the middle? Why can be same process not be applied for them? Name the process used for the extraction of these metals. **Competency Based Que.**

Sol. The metals in the middle of the reactivity series (such as iron, zinc, lead, copper etc.) is moderately reactive.

Thus, to obtain such metals from their compounds, their sulphides and carbonates are first converted into their oxides by the process of roasting and calcination respectively and then the metal oxides are reduced to corresponding metal by using suitable reducing agents such as carbon. [2]

On the other hand, metals which is high up in the rea[ctivity] series (such as sodium, magnesium, calcium, alum[inium] etc.) is very reactive and cannot be obtained fr[om] compound by heating with carbon. Therefore, such m[etals] are obtained by electrolytic reduction of their molten [...]

16. Two ores A and B were taken. On heating, ore A g[ives] CO_2, whereas ore B gives SO_2. What steps will you take to convert them into metals? **NCERT Exem[plar]**

Sol. (i) Ore *A* is a carbonate ore as it gives CO_2 on heating. The steps involved in extraction of *A* are

(a) **Calcination** The carbonate ore is heated strong[ly in] the limited supply of air to produce metal oxide.

$$ACO_3(s) \xrightarrow{\Delta} AO(s) + CO_2(g)$$

(b) **Reduction to metal** The oxide ore is reduced [by] C (coke).

$$AO(s) + C(s) \longrightarrow A(s) + CO(g)$$

(ii) Ore *B* is a sulphide ore as it gives SO_2 on heating.

Following steps are involved in its extraction:

(a) **Roasting** The sulphide ore is heated strongly in the presence of excess of air to produce metal oxide.

$$2BS(s) + 3O_2(g) \xrightarrow{\Delta} 2BO(s) + 2SO_2(g)$$

(b) **Reduction** Oxide of metal *B* is reduced by carbon to obtain the corresponding metal.

$$BO(s) + \underset{\text{Carbon}}{C(s)} \longrightarrow \underset{\text{Metal}}{B(s)} + CO_2/CO(g)$$

17. A metal *X*, which is used in thermit process, when heated with oxygen gives an oxide *Y* which is amphoteric in nature. Identify *X* and *Y*. Write balan[ced] chemical equations of the reactions of oxide *Y* with hydrochloric acid and sodium hydroxide. **CBSE 2[...]**

Sol. Since, the metal '*X*' is used in thermit process, therefore '*X*' is aluminium and '*Y*' is aluminium oxide, A[...] (amphoteric in nature).

$$\underset{(X)}{4Al} + 2O_3 \longrightarrow \underset{(Y)}{2Al_2O_3}$$

Balanced chemical equation is as follow

(i) $$\underset{\text{Aluminium oxide}}{Al_2O_3(s)} + \underset{\text{Hydrochloric acid}}{6HCl(aq)} \longrightarrow \underset{\text{Aluminium chloride}}{2AlCl_3(aq)} + 3H_2O(l)$$

(ii) $$\underset{\text{Aluminium oxide}}{Al_2O_3(s)} + \underset{\text{Sodium hydroxide}}{2NaOH(aq)} \longrightarrow \underset{\text{Sodium aluminate}}{2NaAlO_2(aq)} + H_2O(l)$$

18. Write electronic configuration of sodium (at. no. 1[1]) and oxygen (at. no. 8) and show the formation of t[he] ionic compound obtained when these two elemen[ts] combine. Name anion and cation present in t[he] compound. **CBSE 20[...]**

Sol. The electronic configuration of sodium (Na) and oxygen (O) are as follows

$$\underset{11}{}Na \Rightarrow \overset{K\ L\ M}{2, 8, 1}\ ;\ \underset{8}{}O \Rightarrow \overset{K\ L}{2, 6}$$ [1]

Sodium has only one electron in its outermost shell and oxygen has six electrons in their outermost shell. In order to achieve the nearest noble gas configuration, sodium looses one electron to form sodium ion and oxygen gain 2 electrons to form oxygen ion. When these two combines, they forms ionic bond. It can be represented as

$$2\,\dot{Na} + \ddot{\underset{\times\times}{O}}{:} \longrightarrow \underbrace{[Na^+]_2}_{\text{Cation}} \underbrace{\left[:\ddot{\underset{\times\times}{O}}{:}\right]^{-2}}_{\text{Anion}}$$

[2]

19. (i) What is the significance of the reactivity series of metal?

(ii) Why sodium forms sodium hydroxide when react with water whereas aluminium forms only aluminium oxide?

Sol. (i) The reactivity series of metals is the array of metals in the decreasing order of their reactivity. It helps in predicting wheather a metal can displace another metal or not in a displacement reaction. [1]

(ii) The metals placed lower in the reactivity series are less reactive towards water. Sodium metal placed above aluminium reacts with water to form sodium oxide which further dissolves in water to give sodium hydroxide solution. Whereas, aluminium reacts with oxygen to form aluminium oxide and it does not dissolve in water to form aluminium hydroxide. [2]

20 What happens, when

(i) $ZnCO_3$ is heated in the absence of oxygen?

(ii) a mixture of Cu_2O and Cu_2S is heated?

NCERT Exemplar

Sol. (i) When $ZnCO_3$ is heated in the absence of oxygen then zinc oxide and carbon dioxide are produced.

$$\underset{\text{Zinc carbonate}}{ZnCO_3(s)} \xrightarrow[\text{Calcination}]{\text{Heat}} \underset{\text{Zinc oxide}}{ZnO(s)} + \underset{\text{Carbon dioxide}}{CO_2(g)}$$ [1]

Moisture and volatile impurities are also expelled out during the process of calcination. [1/2]

(ii) When a mixture of Cu_2O and Cu_2S is heated then copper metal and sulphur dioxide gas are produced. [1/2]

$$\underset{\text{Cuprous oxide}}{2Cu_2O(s)} + \underset{\text{Cuprous sulphide}}{Cu_2S(s)} \xrightarrow[\text{Reduction}]{\text{Heat}} \underset{\text{Copper}}{6Cu(s)} + \underset{\text{Sulphur dioxide}}{SO_2(g)}$$ [1]

21. (i) Hydrogen is not a metal but it has been assigned a place in the reactivity series of metals. Explain.

(ii) Name most and least reactive metals.

Sol. (i) Although, hydrogen is not a metal yet it has been assigned a place in the reactivity series of metals. The reason is that like metal, hydrogen also has a tendency to lose electron and forms a positive ion H^+. [1]

The metals which lose electrons less readily than hydrogen are placed below it and the metals which lose electrons more readily than hydrogen are placed above it in the reactivity series of metals. [1]

(ii) Sodium is most reactive and platinum is least reactive metals. [1]

22. During extraction of metals, electrolytic refining is used to obtain pure metals.

(i) Which material will be used as anode and cathode for refining of silver metal by this process?

(ii) Suggest a suitable electrolyte also.

(iii) In this electrolytic cell, where do we get pure silver after passing electric current? **NCERT Exemplar**

Sol. (i) **Anode** Impure block of silver metal.

Cathode Pure thin strip of silver metal. [1]

(ii) Aqueous solution of a silver salt like $AgNO_3$ can be used as an electrolyte. [1]

(iii) We get pure silver at cathode, because at cathode, reduction reaction will take place.

$$Ag^+(aq) + e^- \longrightarrow Ag(s)$$ [1]

23. A teacher asks her students to identify a metal, *M*. She gives them the following clues to help them.

(a) Its oxide reacts with both HCl and NaOH.

(b) It does not react with hot or cold water but reacts with steam.

(c) It can be extracted by electrolysis of its ore.

(i) Identify the metal.

(ii) Write the chemical equations for the reaction of the metal with HCl and NaOH respectively.

(iii) What would happen if the metal is reacted with iron oxide?

Sol. (i) The metal is aluminium because its oxides reacts with both acids and bases to produce salt and water.

(ii) The chemical equations for the reaction of the aluminium with HCl and NaOH is as [1]

$$2Al + 6HCl \longrightarrow 2AlCl_3 + 3H_2$$

$$2Al + 2NaOH + 6H_2O \longrightarrow 2Na[Al(OH)_4] + 3H_2$$

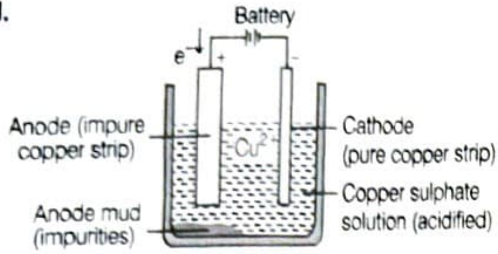
(iii) Reaction of Al with iron oxide takes place as follows:

$$2Al + Fe_2O_3 \longrightarrow Al_2O_3 + 2Fe$$ [1]

Aluminium reacts with iron oxide to form aluminium oxide (Al_2O_3) and molten iron. In this reaction aluminium displaces iron from iron oxide.

24. Explain the process of electrolytic refining for copper with the help of a labelled diagram. **CBSE 2016**

Sol.



[1]

In electrolytic process, the impure metal is made the anode and a thin strip of pure metal is made the cathode. A solution of the metal salt is used as an electrolyte.

On passing the current through the electrolyte, the pure metal from the anode dissolves into the electrolyte.

An equivalent of pure metal from the electrolyte is deposited on the cathode. [1]

At cathode, $Cu^{2+} + 2e^- \longrightarrow Cu$ (deposited)

At anode, $\underset{\text{(Impure metal)}}{Cu(s)} \longrightarrow \underset{\text{(Dissolved)}}{Cu^{2+}(aq) + 2e^-}$ [1]

25. An element *A* reacts with water to form a compound *B* which is used in white washing. The compound *B* on heating forms an oxide *C* which on treatment with water gives back *B*. Identify *A*, *B* and *C* and give the reactions involved. **NCERT Exemplar**

Sol. Element *A* is calcium (Ca) as it reacts with water to form calcium hydroxide. Thus, compound *B* is calcium hydroxide [$Ca(OH)_2$], which is used in white washing.

$$\underset{(A)}{Ca(s)} + 2H_2O(l) \longrightarrow \underset{(B)}{Ca(OH)_2(aq)} + H_2(g) \quad [1]$$

Compound *B* on heating gives CaO.

$$\underset{(B)}{Ca(OH)_2(aq)} \xrightarrow{\Delta} \underset{(C)}{CaO(s)} + H_2O(l)$$

Thus, *C* is calcium oxide (CaO). [1]

(CaO) on treatment with water gives back [$Ca(OH)_2$].

$$\underset{(C)}{CaO(s)} + H_2O(l) \longrightarrow \underset{(B)}{Ca(OH)_2(aq)} \quad [1]$$

26. The thermit process is used for repairing cracks in railway tracks on site.

(i) Write the equation for the reaction taking place in the process, mentioning the physical state of the reactants and products.

(ii) What information in the chemical equation indicates that the reaction is exothermic?

CBSE Additional Practice Question

Sol. Refer to text on Pg-67 (Thermit welding). [3]

27. Write the constituents of solder alloy. Which property of solder makes it suitable for welding electrical wires? **CBSE 2023**

Sol. Refer to text on Pg-68 (Alloy). [3]

Long Answer Type Questions

1. (i) Predict the reaction, if any, between

(a) zinc and silver nitrate solution.

(b) magnesium and iron (II) chloride solution.

(c) copper and magnesium sulphate solution.

Write the equations, with its physical form symbols, for the reaction.

(ii) A lump of element *X* can be cut by a knife. During reaction with water, *X* floats and melts. What is X? Explain. **Competency Based**

Sol. (i) (a) Zinc is more reactive than silver. It will displace from silver nitrate solution.

$$Zn(s) + 2AgNO_3(aq) \longrightarrow Zn(NO_3)_2(aq) + 2Ag$$

(b) Magnesium is more reactive than iron. It will displace iron from iron (II) chloride solution.

$$Mg(s) + FeCl_2(aq) \longrightarrow MgCl_2(aq) + Fe(s)$$

(c) Copper is less reactive than magnesium. It will not displace magnesium from magnesium sulphate solution. Thus, no reaction will take place.

$$Cu(s) + MgSO_4(aq) \longrightarrow \text{No reaction}$$

(ii) *X* is potassium (K). Potassium being soft can be cut with knife and being lighter than water floats at the surface. The heat produced during reaction with water will melt it.

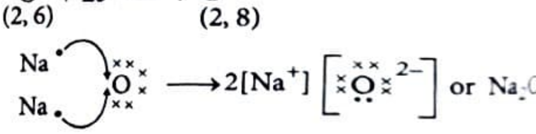
2. (i) With the help of a suitable example, explain how ionic compounds are formed? State any general properties of ionic compounds.

(ii) What happens when aluminium oxide reacts with sodium hydroxide?

Sol. (i) Ionic compounds are formed by the transfer of electrons from one atom to another. These compounds are composed of positively charged metal cations and negatively charged anion, e.g. sodium oxide Na_2O.

$$\underset{(2,8,1)}{2Na} \longrightarrow \underset{(2,8)}{2Na^+} + 2e^-$$

$$\underset{(2,6)}{O} + 2e^- \longrightarrow \underset{(2,8)}{O^{2-}}$$



Properties of ionic compounds are

(a) These compounds are water soluble.

(b) These compounds have high melting and boiling points.

(c) These compounds conduct electricity in molten or in the form of aqueous solution.

(ii) Aluminium oxide (Al_2O_3) reacts with sodium hydroxide solution, it forms sodium aluminate and water.

$$\underset{\text{Aluminium oxide}}{Al_2O_3(s)} + \underset{\text{Sodium hydroxide}}{2\,NaOH(aq)} \longrightarrow \underset{\text{Sodium aluminate}}{2Na\,AlO_2(aq)} + H_2O(l)$$

3. Give reasons for the following:

(i) Ionic compounds have higher melting and boiling points.

(ii) Sodium is kept immersed in kerosene.

(iii) Reaction of calcium with water is less violent.

(iv) Prior to reduction, the metal sulphides and carbonates must be converted into metal oxides for extracting metals. **Competency Based Que.**

Sol. (i) Refer to Q.9 of NCERT Folder (Intext Questions) on Pg-71. [1]

(ii) Refer to Q.3 of NCERT Folder (Intext Questions) on Pg-70. [1½]

(iii) Calcium reacts with water to form calcium hydroxide and hydrogen. The heat produced in this reaction is less which is insufficient to burn the hydrogen gas which is formed. Hence, the reaction of calcium with water is less violent. [1½]

(iv) It is easier to obtain metal from its oxide rather than its carbonates and sulphides. Therefore, these ores are first converted to oxide and then reduced to metal. [1]

4. Explain the following :

(i) Sodium chloride is an ionic compound which does not conduct electricity in solid state whereas it does conduct electricity in molten state as well as in aqueous solution.

(ii) Reactivity of aluminium decrease if it is dipped in nitric acid.

(iii) Metals like calcium and magnesium are never found in their free state in nature. **CBSE 2019**

Sol. (i) Refer to text on Pg-65 (Conduction of electricity). [2]

(ii) When aluminium is dipped in nitric acid, a layer of aluminium oxide is formed on the metal. This happens because nitric acid is a strong oxidising agent. The layer of aluminium oxide prevents further reaction of aluminium. Due to this, the reactivity of aluminium decreases. [2]

(iii) Ca and Mg are alkaline earth metals. They are most reactive metals and readily reacts with atmospheric oxygen and other gases. Therefore, they are found in nature in the form of their compounds. [1]

5. Give reasons for the following:

(i) Generally no hydrogen gas is evolved when metals react with dilute nitric acid.

(ii) Sodium hydroxide solution cannot be kept in aluminium containers.

(iii) Silver metal does not combine easily with oxygen but silver jewellery tarnishes after some time.

(iv) Sodium is obtained by the electrolysis of its molten chloride and not from its aqueous solution.

(v) Aluminium reacts with dilute hydrochloric acid slowly in the beginning.

Sol. (i) Nitric acid (HNO_3) is a strong oxidising agent. The hydrogen gas is formed during the reaction between a metal and dilute nitric acid. The nitric acid oxidises this hydrogen to water and itself gets reduced to NO_2 or NO or N_2O.

So, in the reaction of metals (except Mn and Mg) with dilute nitric acid, no hydrogen gas is evolved. [1]

(ii) Sodium hydroxide reacts with Al thereby corroding the metal and produces highly flammable hydrogen gas. Hence, it cannot be kept in Al containers. [1]

(iii) Silver is a highly unreactive metal. So, it does not react with the oxygen of air easily. But silver jewellery tarnishes after some time (turns black). It is due to the formation of a thin sulphide layer on their surface by the action of hydrogen sulphide (H_2S) gas present in air. [1]

(iv) We cannot use an aqueous solution of sodium chloride to obtain sodium metal. It is because if we electrolyse an aqueous solution of NaCl, then as soon as sodium metal is produced at cathode, it will react with water present in the aqueous solution to form sodium hydroxide. So, the electrolysis of aqueous solution of NaCl will produce NaOH instead of Na metal. [1]

(v) Al metal reacts with dilute HCl slowly in the beginning due to the presence of a tough protective layer of aluminium oxide on its surface. [1]

6. On the basis of reactivity, metals are grouped into three categories **CBSE 2023**

(i) Metals of low reactivity

(ii) Metals of medium reactivity

(iii) Metals of high reactivity

Therefore metals are extracted in pure form from their ores on the basis of their chemical properties.

Metals of high reactivity are extracted from their ores by electrolysis of the molten ore.

Metals of low reactivity are extracted from their sulphide ores, which are converted into their oxides. The oxides of these metals are reduced to metals by simple heating.

(a) Name the process of reduction used for a metal that gives vigorous reaction with air and water both.

(b) Carbon cannot be used as a reducing agent to obtain aluminium from its oxide? Why?

(c) Describe briefly the method to obtain mercury from cinnabar. Write the chemical equation for the reactions involved in the process.

Sol. (a) Refer to text on Pg-67 (Electrolytic reduction). [2]

(b) Al cannot be obtained from aluminium oxide with carbon (as reducing agent) as it is more reactive than carbon and so can't be displaced by carbon. [1]

(c) Refer to text on Pg-66 (Extraction of metal of low reactivity). [2]

7. (i) Write chemical equations for the following reactions :

(a) Calcium metal reacts with water.

(b) Cinnabar is heated in the presence of air.

(c) Manganese dioxide is heated with aluminium powder.

(ii) What are alloys ? List two properties of alloys. **CBSE 2019**

Sol. (i) (a) Refer to text on Pg-62-63 (Reaction of metals with water). [1]

(b) Refer to text on Pg-66 (Extraction of metals of low reactivity). [1]

(c) As aluminium is more reactive than manganese, so it displace manganese from its oxide. The following reaction takes place.

$$3MnO_2 + 4Al \longrightarrow 3Mn + 2Al_2O_3 \quad [1]$$

(ii) Refer to text on Pg-68 (Alloy). [2]

8. (i) Given below are the steps for extraction of copper from its ore. Write the reaction involved.

(a) Roasting of copper (I) sulphide.

(b) Reduction of copper (I) oxide with copper (I) sulphide.

(c) Electrolytic refining.

(ii) Draw a neat and well labelled diagram for electrolytic refining of copper. **NCERT Exemplar**

Sol. (i) (a) $2Cu_2S(s) + 3O_2(g) \xrightarrow{\text{Heat}} 2Cu_2O(s) + 2SO_2(g)$ [1]

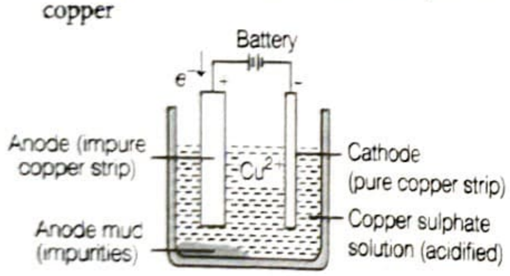
(b) $2Cu_2O(s) + Cu_2S(s) \xrightarrow{\text{Heat}} 6Cu(s) + SO_2(g)$ [1]

(c) Copper is refined electrolytically. The reactions are as follows

At cathode, $Cu^{2+}(aq) + 2e^- \longrightarrow Cu(s)$

At anode, $Cu(s) \longrightarrow Cu^{2+}(aq) + 2e^-$ [1½]

(ii) The diagram shown below is of electrolytic refining of copper



[1½]

9. (i) Give differences between roasting and calcination with suitable examples.

(ii) Explain how the following metals are obtained from their compounds by the reduction process.

(a) Metal *M* which is in the middle of the reactivity series.

(b) Metal *N* which is high up in the reactivity series.

Give one example of each type.

Sol. (i)

Roasting	Calcination
• Ore is heated in excess of air.	Ore is heated in the absence or limited supply of air.
• This is used for sulphide ores.	This is used for carbonate
• SO_2 is produced along with metal oxide.	CO_2 is produced along with metal oxide.
• e.g. $2ZnS(s) + 3O_2(g) \xrightarrow{\Delta} 2ZnO(s) + 2SO_2(g)$	e.g. $ZnCO_3(s) \xrightarrow{\Delta} ZnO(s) + CO$

(ii) (a) The metal *M* which is in the middle of the reactivity series (such as iron, zinc, lead, copper etc.) is moderately reactive.

Thus, to obtain such metals from their compounds their sulphides and carbonates (in which they are present in nature) are first converted into their oxides by the process of roasting and calcination respectively.

The metal oxides (*M*O) are then reduced to the corresponding metals by using suitable reducing agents such as carbon, e.g. zinc metal can be obtained from its oxide as follows

$$\underset{\text{Zinc oxide}}{ZnO(s)} + \underset{\text{Carbon}}{C(s)} \longrightarrow \underset{\text{Zinc}}{Zn(s)} + \underset{\text{Carbon monoxide}}{CO(g)}$$

(b) The metal *N* which is high up in the reactivity series (such as sodium, magnesium, calcium, aluminium) is very reactive and cannot be obtained from compound by heating with carbon.

Therefore, such metals are obtained by electrolytic reduction of their molten salt, e.g. sodium is obtained by the electrolysis of molten sodium chloride (NaCl)

At cathode, $Na^+ + e^- \longrightarrow Na$

At anode, $2Cl^- \longrightarrow Cl_2 + 2e^-$

10. In what forms metals are found in nature? With the help of examples, explain how metals react with oxygen, water and dilute acids. Also, write chemical equations for the reactions.

Sol. Metals are found in both free and combined states.

Reaction of metals with oxygen All metals combine with oxygen at different rates to form metal oxides, e.g. sodium forms sodium oxide at room temperature on reacting with oxygen.

$$4Na + O_2 \longrightarrow 2Na_2O$$

But copper forms copper oxide when it is heated in presence of air.

$$2Cu + O_2 \xrightarrow{\Delta} 2CuO$$

Reaction of metals with water Metals like sodium react with cold water, magnesium with hot water to form hydroxides and evolve hydrogen gas. But iron reacts with steam to form oxides and evolve hydrogen gas.

$$2Na(s) + 2H_2O(l) \longrightarrow 2NaOH(aq) + H_2(g)$$
$$Mg(s) + 2H_2O(l) \longrightarrow Mg(OH)_2(aq) + H_2(g)$$
$$3Fe(l) + 4H_2O(l) \longrightarrow Fe_3O_4(s) + 4H_2(g)$$ [1½]

Reaction of metals with acids Metals react with acids to form salt and evolve hydrogen gas.

e.g. Magnesium reacts with dilute hydrochloric acid to form magnesium chloride and evolve H_2.

$$Mg(s) + 2HCl(aq) \longrightarrow MgCl_2(aq) + H_2(g)\uparrow$$ [1½]

11. (i) Write the steps involved in the extraction of pure metals in the middle of the activity series from their carbonate ores.

(ii) How is copper extracted from its sulphide ore? Explain the various steps supported by chemical equations. Draw labelled diagram for the electrolytic refining of copper.

Sol. (i) Steps involved in the extraction of pure metals in the middle of the activity series from their carbonate ores.

Carbonate ore
↓
Calcination [1]
(Carbonate ores are changed into oxides by heating strongly in limited supply of air)
↓
Reduction of metal
(Metal oxides are reduced to the corresponding metals by using siutable reducing agent) [1]
↓
Purification of metal
(Removal of impurities) [1]

(ii) Refer to text on Pg-66 and 67-68 (Copper glance and electrolytic refining) [2]

12. Two ores *X* and *Y* were taken. On heating these ores, it was observed that

(i) ore *X* gives CO_2 gas, and

(ii) ore *Y* gives SO_2 gas.

Write steps to convert these ores into metals, giving chemical equations of the reactions that take place.

CBSE 2020

Sol. (i) The ore *X* which gives CO_2 gas is a carbonate ore. [1/2]

(ii) The ore *Y* which gives SO_2 gas is a sulphide ore. [1/2]

The steps involves in the conversion of these ores into metals is given as below:

(a) **Enrichment of ores** Removal of unwanted material (gangue) from the ore is called enrichment or concentration of ore.
The undesirable impurities like soil, sand etc., are gangue or matrix. [1]

(b) **Conversion of ores into oxides**

$$\underset{(X)}{ZnCO_3} \xrightarrow{\text{Calcination}} ZnO + CO_2\uparrow$$

$$\underset{(Y)}{2ZnS} + 3O_2 \xrightarrow{\text{Roasting}} 2ZnO + 2SO_2\uparrow$$ [1]

(c) Reaction of oxide ore (smelting) is as follows

$$\underset{\text{Zinc oxide}}{ZnO} + \underset{\text{Carbon}}{C} \longrightarrow \underset{\text{Zinc}}{Zn} + \underset{\text{Carbon monoxide}}{CO}$$ [1]

(d) **Refining** It is the process of purification of the metal obtained after reduction. Various methods for refining are employed but most common method is electrolytic refining. [1]

13. (i) With the help of a diagram explain the method of refining of copper by electrolysis.

(ii) How are broken railway tracks joined? Give the name of the process and the chemical equation of the reaction involved. **CBSE 2020**

Sol. (i) Refer to text on Pg-67 (Refining of metals). [2½]

(ii) The broken railway tracks are joined by the reaction of iron (III) oxide with aluminium. This process is called thermite welding. The chemical equation involved is given as follows

$$\underset{\text{Ferric oxide}}{Fe_2O_3(s)} + \underset{\text{Aluminium}}{2Al(s)} \longrightarrow \underset{\text{Iron}}{2Fe(l)} + \underset{\text{Aluminium oxide}}{Al_2O_3(s)} + \text{Heat}$$ [2½]

ASSESSMENT EXERCISE

Competency Focused (Objective) Questions

Multiple Type Questions

1 An element *A* is soft and can be cut with a knife. This is very reactive to air and cannot be kept open in air. It reacts vigorously with water. Identify the element from the following.

(a) Mg (b) Na
(c) P (d) Ca

2 Which one of the following four metals would be displaced from the solution of its salts by other three metals?

(a) Mg (b) Ag
(c) Zn (d) Cu

3 When a non-metal is allowed to react with water,

(a) CO_2 gas is formed.
(b) H_2 gas is formed.
(c) product formed depends on the temperature.
(d) no products are formed.

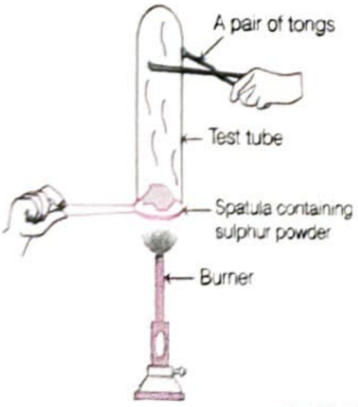
4 During electrolytic refining of zinc, it gets

(a) deposited on cathode
(b) deposited on anode
(c) deposited on cathode as well as anode
(d) remains in the solution

5 Which of the following can udnergo a chemical reaction?

(a) $MgSO_4 + Fe$
(b) $ZnSO_4 + Fe$
(c) $MgSO_4 + Pb$
(d) $CuSO_4 + Fe$

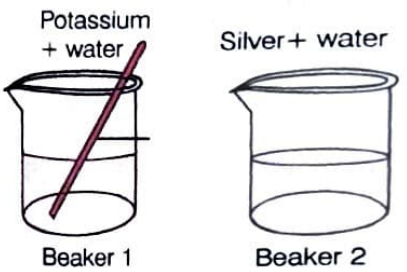
6 Pratyush took sulphur powder on a spatula and heated it. He collected the gas evolved by inverting a test tube over it as shown in figure. The gas evolved is



Competency Based Que.

(a) O_2 (b) SO_2
(c) CO_2 (d) H_2 and SO_2

7 A student drops pieces of potassium and silver in beakers containing water. The image shows the rea

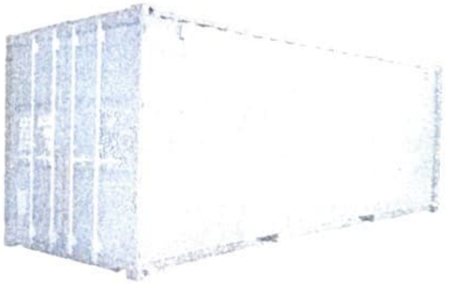


What are the products formed in each beaker?

(a) Beaker 1 : K_2O and H_2O ; Beaker 2 : No reaction takes
(b) Beaker 1 : KOH and H_2 ; Beaker 2 : Ag_2O and H_2O
(c) Beaker 1 : K_2O and H_2O ; Beaker 2 : AgO and H_2O
(d) Beaker 1 : KOH and H_2 ; Beaker 2 : No reaction takes

8 Shown below is a container that is used in the transportation of goods over long distances.

Competency Base



These containers are made of steel. Which property steel is mainly used to make these containers?

(a) Its ductility
(b) Its malleability
(c) Its metallic lustre
(d) Its electrical conductivity

Assertion-Reason Questions

Direction (Q. Nos. 9-11) *In each of the following questio a statement of Assertion (A) is given by the corresponding statement of Reason (R). Of the statements, mark the correct answer as*

(a) Both Assertion and Reason are true and Reason is the correct explanation of Assertion.
(b) Both Assertion and Reason are true, but Reason is no the correct explanation of Assertion.
(c) Assertion is true, but Reason is false.
(d) Assertion is false, but Reason is true.

9 **Assertion** (A) Hydrogen is not a metal but it has been assigned a place in the reactivity series of metals.

Reason (R) Hydrogen has a tendency to lose electron and forms a positive ion H^+.

10 **Assertion** (A) When lead vessel is kept in a solution of $ZnSO_4$, lead will remain unaffected.

Reason (R) Lead is less reactive than zinc.

11 **Assertion** (A) Zinc becomes dull in moist air.

Reason (R) Zinc is coated by a thin film of its basic carbonate in moist air.

Answers

1. (b) **2.** (b) **3.** (d) **4.** (a) **5.** (d)

6. (b) The gas evolved is SO_2.

$$\underset{\text{Sulphur}}{S(s)} + \underset{\text{Oxygen}}{O_2(g)} \longrightarrow \underset{\text{Sulphur dioxide}}{SO_2(g)}$$

7. (d) KOH is formed in beaker 1, while no reaction takes place in beaker 2 as silver has low reactivity than H. Thus, it can not replace H from water.

$$2K + 2H_2O \longrightarrow 2KOH + H_2$$

8. (b) Malleability property of metals (steel) is mainly used to make these containers.

9. (a) **10.** (a) **11.** (a)

Case Based/Source Based Questions

12 **Direction** [Q.Nos. (i)-(iii)] ***Answer the questions on the basis of your understanding of the following passage, table and related studied concepts:***

Generally, the metals form basic oxides whereas non-metals form acidic oxides. Only a few metals form hydrides with hydrogen and these hydrides are unstable while the non-metals form stable hydrides.

Metal atom can donate electrons easily hence they act as reducing agents and the non-metals can accept electrons readily, therefore they act as oxidising agent. Thermal conductivity is the ability of an element to allow heat to pass through it.

Generally, all metals are good conductors of heat except lead and mercury. The table shows the thermal conductivity values of some elements expressed in Watt/centimeter Kelvin (W/cm-K).

Element	Symbol	Thermal conductivity (W/cm-K)
Lead	Pb	0.34
Copper	Cu	4.01
Aluminium	Al	2.37
Iron	Fe	0.802
Selenium	Se	0.0204
Sulphur	S	0.00269
Phosphorus	P	0.00235

(i) Name a metal which is a best conductor of heat and name the electrolyte used in refining of copper.

(ii) Which reducing agent can be used in the extraction of metals placed at the top of the reactivity series?

(iii) Which metal form both acidic oxide and basic oxide. Classify Na_2O, Al_2O_3 and SO_3 as acidic, basic or amphoteric oxide.

Or

(iii) An element forms an oxide A_2O_3 which is acidic in nature. Identify A is metal or non-metal.

Constructed Response (Descriptive) Questions

Very Short Answer Type Questions

13 Which of the following elements is a metal? Explain.

$${}^{7}_{3}X,\ {}^{3}_{1}Y,\ {}^{10}_{9}Z$$

14 (i) Name one metal and one non-metal which are obtained on a large scale from sea water.

(ii) Name the process used for the concentration of the sulphide ore.

15 Reverse of the following chemical reaction is not possible.

$$Zn(s) + CuSO_4(aq) \longrightarrow ZnSO_4(aq) + Cu(s)$$

Justify this statement.

16 For the reduction of metal oxide to metal, suggest a reducing agent cheaper than aluminium.

17 An ore, on heating in air, give sulphur dioxide gas. Name the method in each metallurgical step, that will be required to extract this metal from its ore.

Short Answer Type Questions

18 Answer the following:

(i) Name a metal which does not stick to glass.

(ii) What is the nature of zinc oxide?

(iii) What is deposited at the cathode, a pure or impure metal?

(iv) Will carbon monoxide (CO) change the colour of blue litmus?

19 Zn is more reactive than Fe. Therefore, it should get corroded faster than Fe. But it does not happen, instead, it is used to galvanise iron. Explain why does it happen so?

Competency Based Que.

20 Carbon cannot reduce the oxides of sodium, magnesium and aluminium to their respective metals. Why? Where are these metals placed in the reactivity series. How are these metals obtained from their ores?

Competency Based Que.

21 A solution of $CuSO_4$ was kept in an iron pot. After few days, the iron pot was found to have a number of holes in it. Explain the reason in terms of reactivity. Write the equation of the reaction involved. **Competency Based Que.**

22 State which of the following reactions will take place or which will not, giving suitable reason for each?

$$Zn(s) + CuSO_4(aq) \longrightarrow ZnSO_4(aq) + Cu(s)$$
$$Fe(s) + ZnSO_4(aq) \longrightarrow FeSO_4(aq) + Zn(s)$$

Long Answer Type Questions

23 A metal '*M*' is stored under kerosene. It vigorously catches fire, if a small piece of this metal is kept open in air. Dissolution of this metal in water releases great amount of energy and the metal catches fire. The solution so formed turns red litmus blue.

(a) Name the metal '*M*'.

(b) Write formula of the compound formed when this metal is exposed to air.

(c) Why is metal '*M*' stored under kerosene?

(d) If oxide of this metal is treated with hydro what would be the products?

(e) Write balanced equations for
(i) Reaction of '*M*' with air.
(ii) Reaction of '*M*' with water.
(iii) Reaction of metal oxide with hydrochlor

24 Three pieces of a rust free iron are completely with the following: **Competency**

(i) Plastic (ii) Oil paint
(iii) Zinc

An identical scratch is made on each piece, thu exposing the iron. The pieces of iron are kept e moist air for 10 days and then checked for rust

(a) State if rusting will be observed at the p scratch on the three iron peices.

(b) Give reasons for your answer in each case.

(c) Name the process of applying a protective z to steel or iron.

CHAPTER 04

Carbon and Its Compounds

Carbon is the find most important element after oxygen and hydrogen, in the existence of life on earth. The name 'carbon' arise from Latin word *carbo* which means coal. The earth's crust has only 0.02% carbon which is present in the form of minerals (like carbonates, hydrogen-carbonates, coal and petroleum) and the atmosphere has 0.03% of carbon dioxide.

Fuels, clothing material, paper, rubber, plastics, leather, drugs and dyes are all made up of carbon. Also, all living structures are carbon based. In this chapter, we will study the properties of carbon and its compounds.

4.1 Covalent Bonding in Carbon Compounds

Carbon is a versatile element and a non-metal, which is represented by symbol 'C'.

Atomic number of carbon is 6. So, its electronic configuration $= \overset{K}{2}, \overset{L}{4}$

Thus, there are 4 electrons in its outermost shell and it can attain nearest noble gas configuration by the following two ways

- It could gain 4 electrons and form C^{4-} anion. But for a nucleus having 6 protons, it would be difficult to hold on to 10 electrons, i.e. 4 extra electrons.
- It could lose 4 electrons and form C^{4+} cation. But a large amount of energy is required to remove 4 electrons leaving behind a carbon cation with 6 protons in its nucleus holding on to just two electrons together, which is not possible.

Therefore, in order to overcome this problem, carbon shares its valence electrons with other atoms of carbon or with atoms of other elements. These shared electrons belong to the outermost shells of both atoms and in this way, both atoms attain the nearest noble gas configuration. This type of bonding is called **covalent bonding.**

Thus, covalent bond can be defined as "the bonds which are formed by the sharing of an electron pair between the atoms (either same or different)".

The compounds containing covalent bonds are called **covalent compounds.**

The number of electrons shared between two atoms to complete their octet (except hydrogen which shows duplet) is known as the **covalency** of that atom. Thus, the covalency of hydrogen is 1, oxygen is 2, nitrogen is 3 and carbon is 4.

Lewis Dot Structures

These structures depict the electronic structures of the elements and how the electrons are paired. Each dot or cross represents an electron. A pair of dots between chemical symbols for atoms represents a bond.

Steps for writing the Lewis dot structures of a covalent compound

- Write the electronic configuration of all the atoms present in the molecule.
- Identify the number of electrons required by each atom to complete its octet to achieve the nearest noble gas configuration.
- Share the electrons of each atom with one another in such a manner that all atoms in a molecule have noble gas configuration.
- The shared electrons are counted in the valence shell of both the atoms sharing it.

Some Examples Depicting Covalent Bonding

1. Formation of Hydrogen Molecule (H_2)

Atomic number of H = 1

Electronic configuration $= \overset{K}{1}$

It has 1 electron in its *K*-shell and needs 1 more electron to fill the *K*-shell completely. Thus, two H-atoms share their electrons to form a molecule of H_2. This allows each